

IL-17A-secreting $\gamma\delta$ T cells promote resistance to CDK4/CDK6 inhibitors in HR⁺HER2⁻ breast cancer via CX3CR1⁺ macrophages

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Resistance to cyclin-dependent kinase 4/6 (CDK4/CDK6) inhibitors leads to treatment failure and disease progression in women with hormone receptor⁺HER2⁻ (HR⁺HER2⁻) breast cancer (BC). We delineated a hypoxia-sensitive, CCL2-dependent pathway recruiting interleukin-17A (IL-17A)-secreting $\gamma\delta$ T cells to mouse HR⁺HER2⁻ BCs following CDK4/CDK6 inhibition, resulting in repolarization of tumor-associated macrophages (TAMs) toward an immunosuppressive CX3CR1⁺ phenotype associated with resistance. Increased IL-17A signaling and intratumoral $\gamma\delta$ T cell abundance positively correlated with advanced grade and/or reduced survival in two cohorts of individuals with HR⁺HER2⁻ BC. Circulating $\gamma\delta$ T cells and plasma CCL2 levels negatively correlated with progression in an independent series of individuals with HR⁺HER2⁻ BC receiving CDK4/CDK6 inhibitors. Intratumoral $\gamma\delta$ T cells were increased in post- versus pretreatment biopsies from individuals with HR⁺HER2⁻ BC relapsing on CDK4/CDK6 inhibitors. CX3CR1⁺ TAMs had negative prognostic impact in women with HR⁺HER2⁻ BC receiving neoadjuvant PD-1 blockage and radiotherapy. Thus, $\gamma\delta$ T cells and CX3CR1⁺ TAMs may favor resistance to CDK4/CDK6 inhibitors in individuals with HR⁺HER2⁻ BC.

Three cyclin-dependent kinase 4/6 (CDK4/CDK6) inhibitors are approved for clinical use in combination with endocrine therapy as first-line treatment for advanced/metastatic hormone receptor⁺HER2⁻ (HR⁺HER2⁻) breast cancer (BC), reflecting their established efficacy in this patient population¹. Two of these drugs, abemaciclib and ribociclib, have also demonstrated promising activity when added to adjuvant endocrine therapy in women with high-risk early-stage HR⁺HER2⁻ BC². Despite this progress, many women with late-stage HR⁺HER2⁻ BC receiving CDK4/CDK6 inhibitors ultimately progress and succumb to their disease, owing to hitherto poorly characterized mechanisms of resistance¹. CDK4/CDK6 inhibitors suppress the proliferation of cancer cells with hyperactive CDK4 or CDK6 by preventing the CDK4/CDK6-catalyzed inactivating phosphorylation of RB transcriptional corepressor 1 (RB1), which is required for proliferating cells to enter S phase¹. Accordingly, loss of RB1 activity and other alterations compromising the ability of

malignant cells to arrest at the G1–S checkpoint, such as *TP53* mutations, have been associated with resistance to CDK4/CDK6 inhibitors in preclinical and clinical settings^{3–6}.

Accumulating evidence indicates that CDK4/CDK6 inhibitors also mediate immunostimulatory effects that may contribute to efficacy^{7,8}. For instance, the prototypic CDK4/CDK6 inhibitor palbociclib (P) has been shown to promote the secretion of chemotactic and immunostimulatory cytokines such as C-C motif chemokine ligand 5 (CCL5), C-X-C motif chemokine ligand 9 (CXCL9) and CXCL10 in mouse BC models⁹, at least partially downstream of a permanent proliferative arrest known as cellular senescence¹⁰. Moreover, CDK4/CDK6 inhibitors deplete CD4⁺CD25⁺FOXP3⁺ regulatory T (T_{reg}) cells^{11,12}, which are known to promote a prometastatic tumor microenvironment (TME) in a variety of malignancies¹³. Besides pointing to CDK4/CDK6 inhibitors as promising combinatorial partners for immune checkpoint inhibitors

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(ICIs)^{14,15} and other immunostimulatory strategies such as radiation therapy (RT)^{16–18}, these observations suggest that hitherto unidentified immunological mechanisms may favor resistance to CDK4/CDK6 inhibitors. In vitro and in silico data supporting this possibility have begun to emerge^{19,20}.

We harnessed an immunocompetent mouse model that closely recapitulates the immunobiology of human HR⁺HER2[−] BC²¹ to demonstrate that interleukin-17 (IL-17)-producing $\gamma\delta$ T cells recruited to the TME following CDK4/CDK6 inhibition through a CCL2-dependent mechanism mediate immunosuppressive effects via C-X3-C motif chemokine receptor 1 (CX3CR1)-expressing tumor-associated macrophages (TAMs), ultimately favoring treatment resistance. This pathway can be averted by focal RT delivered before CDK4/CDK6 inhibition, reflecting at least partially the inability of hypoxic tumor areas (which are initially enriched by RT owing to their intrinsic radioresistance)²² to secrete CCL2 following CDK4/CDK6 inhibition and to sustain $\gamma\delta$ T cell survival. In silico data from a public repository as well as immunohistochemistry, multispectral immunofluorescence and single-cell RNA sequencing (scRNA-seq) results from tumor biopsies and data on circulating cytokines and peripheral immune cells from five independent cohorts of individuals with HR⁺HER2[−] BC (including longitudinal samples obtained before and after CDK4/CDK6 or PD-1 inhibition followed by PD-1 inhibition plus RT) suggest that this mechanism may also be operational in humans. Collectively, our findings delineate an immunological mechanism of resistance to CDK4/CDK6 inhibitors that (because IL-17 blockers²³ as well as TAM repolarizing agents²⁴ are licensed for use in humans) may be clinically actionable to improve disease outcome in individuals with HR⁺HER2[−] BC.

Results

CDK4/CDK6 inhibitors recruit $\gamma\delta$ T cells to mouse HR⁺HER2[−] BCs

Given that focal RT improves the therapeutic effects of CDK4/CDK6 inhibition with P when delivered before (RT $\rightarrow$ P) (but not after (P $\rightarrow$ RT)) P¹⁶, we hypothesized that immunological mechanisms would underlie, at least partially, such a sequencing-dependent increase in efficacy. To test this possibility, we harnessed endogenous HR⁺HER2[−] mammary carcinomas established in immunocompetent female C57BL/6 mice by medroxyprogesterone acetate (MPA, M) pellets plus oral 7,12-dimethylbenz[a]anthracene (DMBA, D), a model of multifocal, metachronous HR⁺ carcinogenesis^{21,25}. Besides resembling human HR⁺HER2[−] BC from transcriptional and immunological perspectives²¹ and modeling carcinogenesis in the context of evaded immunosurveillance (at odds with transplantable mouse tumors)²⁶, M/D-driven neoplasms are poorly sensitive to ICIs²¹ but highly responsive to hypofractionated RT and CDK4/CDK6 inhibition¹⁶, recapitulating key therapeutic features of their human counterparts^{1,27}.

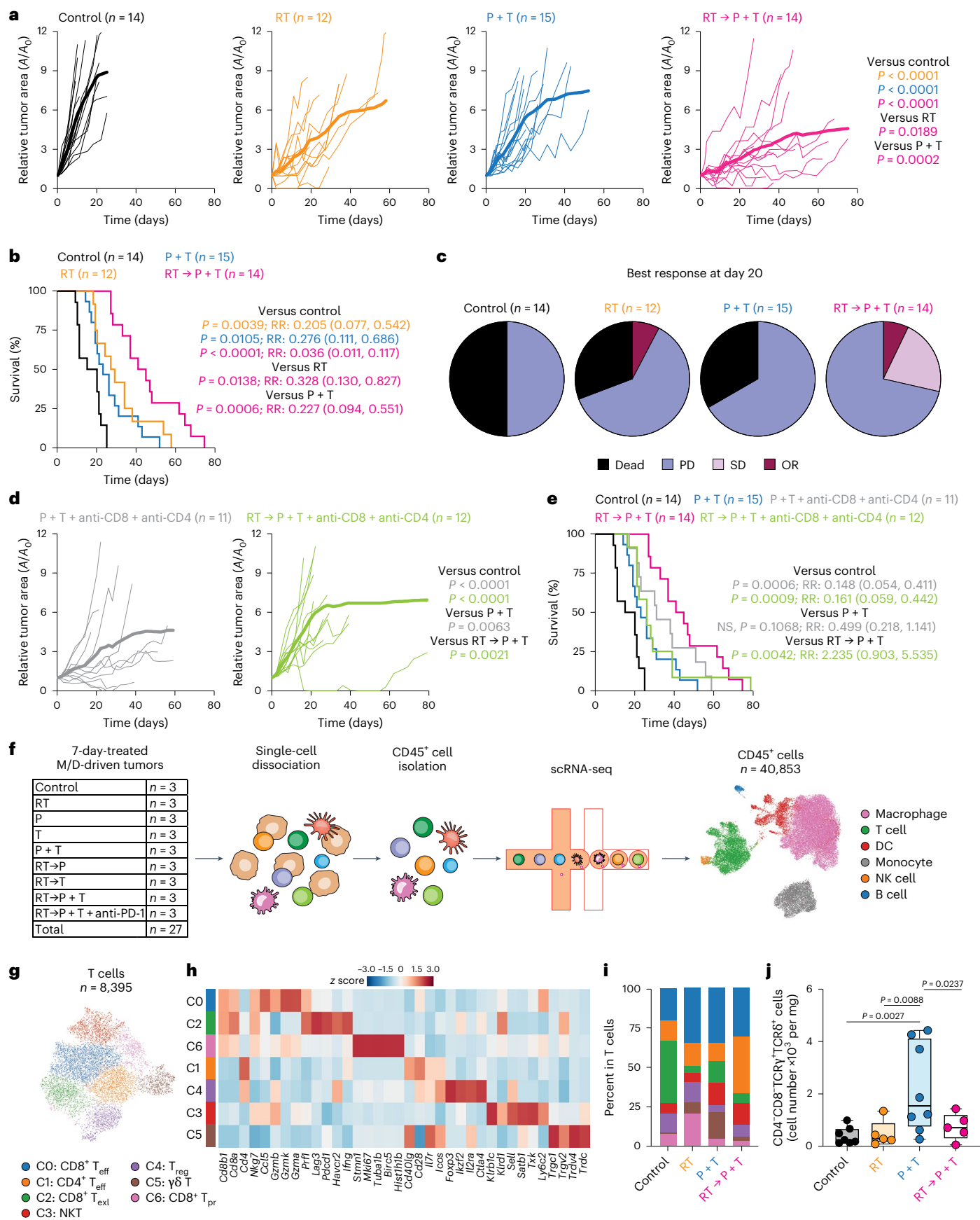
As previously demonstrated for P alone^{16,28}, mice bearing M/D-driven carcinomas responded to oral P combined with tamoxifen (T)-based endocrine therapy (an approach that recapitulates the clinical management of advanced/metastatic HR⁺HER2[−] BC) with robust tumor control at the primary disease site (that is, the first neoplastic lesion to emerge; Fig. 1a and Extended Data Fig. 1a) and a pronounced overall survival (OS) benefit (as dictated by tumor burden at primary and secondary disease sites; Fig. 1b and Extended Data Fig. 1b), an effect that was considerably extended when P + T were preceded by focal RT delivered in three daily fractions of 10 Gy each (Fig. 1a,b and Extended Data Fig. 1a,b). These findings were corroborated by a mouse-adapted version of the Response Evaluation Criteria in Solid Tumors (RECIST) analysis²⁹, demonstrating that mice with M/D-driven tumors exposed to RT $\rightarrow$ P + T were all alive 20 days after treatment initiation and exhibited a >25% disease control (objective responses and stable disease) rate (Fig. 1c and Extended Data Fig. 1c).

Although the depletion of CD4⁺ and CD8⁺ T cells failed to limit (and de facto paradoxically improved) the control of primary (first-appearing) M/D-driven tumors by P + T (Fig. 1d), this did not translate into alterations in OS (Fig. 1e), reflecting no differences in systemic tumor control (Extended Data Fig. 1d). This is at odds with previous data from mouse models of melanoma^{30,31} and HER2⁺ BC⁹, suggesting that the efficacy of CDK4/CDK6 inhibitors involves adaptive immunity. Conversely, CD4⁺ and CD8⁺ T cell depletion limited both the local and systemic efficacy of RT $\rightarrow$ P + T (Fig. 1d,e). These findings delineated a scenario of limited T cell activation by P + T that could be at least in part reversed by focal RT. Although we reasoned that the ability of RT to reverse immunosuppression as driven by P + T could be exacerbated with additional immunostimulatory agents, an ICI specific for PD-1 failed to increase the survival benefits afforded by RT $\rightarrow$ P + T (Extended Data Fig. 1e,f).

To explain these findings, we performed a granular characterization of immune cell populations elicited by RT, P + T and their combination by isolating CD45⁺ cells from M/D-driven carcinomas optionally exposed to these therapeutic regimens (three tumors each) and subjecting them to scRNA-seq (Fig. 1f). By merging a total of 40,853 CD45⁺ cells from 27 independent samples, we identified three lymphoid (T, B and natural killer (NK) cells) and three myeloid (monocyte, macrophages and dendritic cells (DCs)) cell clusters (Fig. 1f and Extended Data Fig. 1g). Reclustering of 8,395 CD3⁺ T cells led to the identification of seven subclusters, including (1) three subclusters with abundant expression of *Cd8b1* and *Cd8a* plus genes involved in effector functions (*Nkg7* and *Ccl5*) and cytotoxicity (*Gzmb*, *Gzmk* and *Gzma*; C0; CD8⁺ effector T (T_{eff}) cells), genes encoding co-inhibitory receptors (*Lag3*, *Pdcd1* and *Havcr2*) and effector (*Nkg7* and *Ifng*) and cytotoxic (*Prf1*) molecules (C2; CD8⁺ exhausted-like T (T_{exl}) cells) and genes associated with T cell proliferation (*Stmn1*, *Mki67*, *Tuba1b*, *Birc5* and *Hist1h1b*; C6;

Fig. 1 | CDK4/CDK6 inhibition recruits $\gamma\delta$ T cells to M/D-driven mammary carcinomas. a–e, C57BL/6 mice bearing M/D-driven tumors were randomly allocated to receive no treatment (control), focal RT, oral P plus T or RT followed by (P + T, $\pm$ anti-CD4 and anti-CD8). Individual growth curves for primary tumors shown as tumor area relative to the area at treatment start (A/A_0 ; **a** and **d**), survival curves (**b** and **e**) and pie charts with best responses at day 20 (**c**) are reported. Pie charts show the percentage of dead mice and mice with objective response (OR; $A/A_0 \leq 0.756$), stable disease (SD; $0.756 < A/A_0 \leq 1.440$) and progressive disease (PD; $A/A_0 > 1.440$), according to RECIST-based criteria. Number of mice, risk ratio (RR) with 95% confidence interval and *P* values are shown. **f**, Schematic representation of CD45⁺ cell collection for scRNA-seq from M/D-driven tumors treated as indicated in the table on days 0–5 and collected at day 6 and uniform manifold approximation and projection (UMAP) visualization of 40,853 cells (pooled from 27 tumors, with three independent biological samples per treatment group each obtained from individual mice), color-coded for the indicated cell type; NK cell, natural killer cell. **g**, UMAP subclustering of T cells in seven phenotypes, as indicated by the color-coded legend

(T_{pr}, proliferating T cells). **h**, Heat map of cluster-defining genes for the seven T cell phenotypes. **i**, Bar plots showing the frequency (average of three independent biological samples per treatment group each obtained from individual mice) of each T cell subcluster in the indicated treatment groups. **j**, Density of CD4⁺CD8⁺TCR γ TCR δ ⁺ ($\gamma\delta$) T cells infiltrating M/D-driven tumors treated with RT (*n* = 5), P + T (*n* = 8) or RT $\rightarrow$ P + T (*n* = 5) or left untreated (control, *n* = 7) and collected at the experimental endpoint. Box plots show median $\pm$ interquartile range; whiskers indicate minima and maxima. Significant (*P* < 0.05) statistical differences are shown. *P* values were determined by a linear mixed effects model followed by simultaneous tests of general linear hypotheses (**a** and **d**), two-sided Gehan–Breslow–Wilcoxon test (**b** and **e**) and one-way analysis of variance (ANOVA) plus Fisher's least significant difference (LSD) test (**j**). For data shown in **i**, relevant FDR-corrected *P* values from a full pairwise χ^2 statistical test between treatments for each cluster are reported in the main text. Part of the results shown in **b** are also depicted in **e** (control, P + T and RT $\rightarrow$ P + T), Figs. 2d, 3f and 4i (control and P + T) and Extended Data Fig. 1b (control) and Extended Data Fig. 1f (control and RT $\rightarrow$ P + T); NS, not significant.



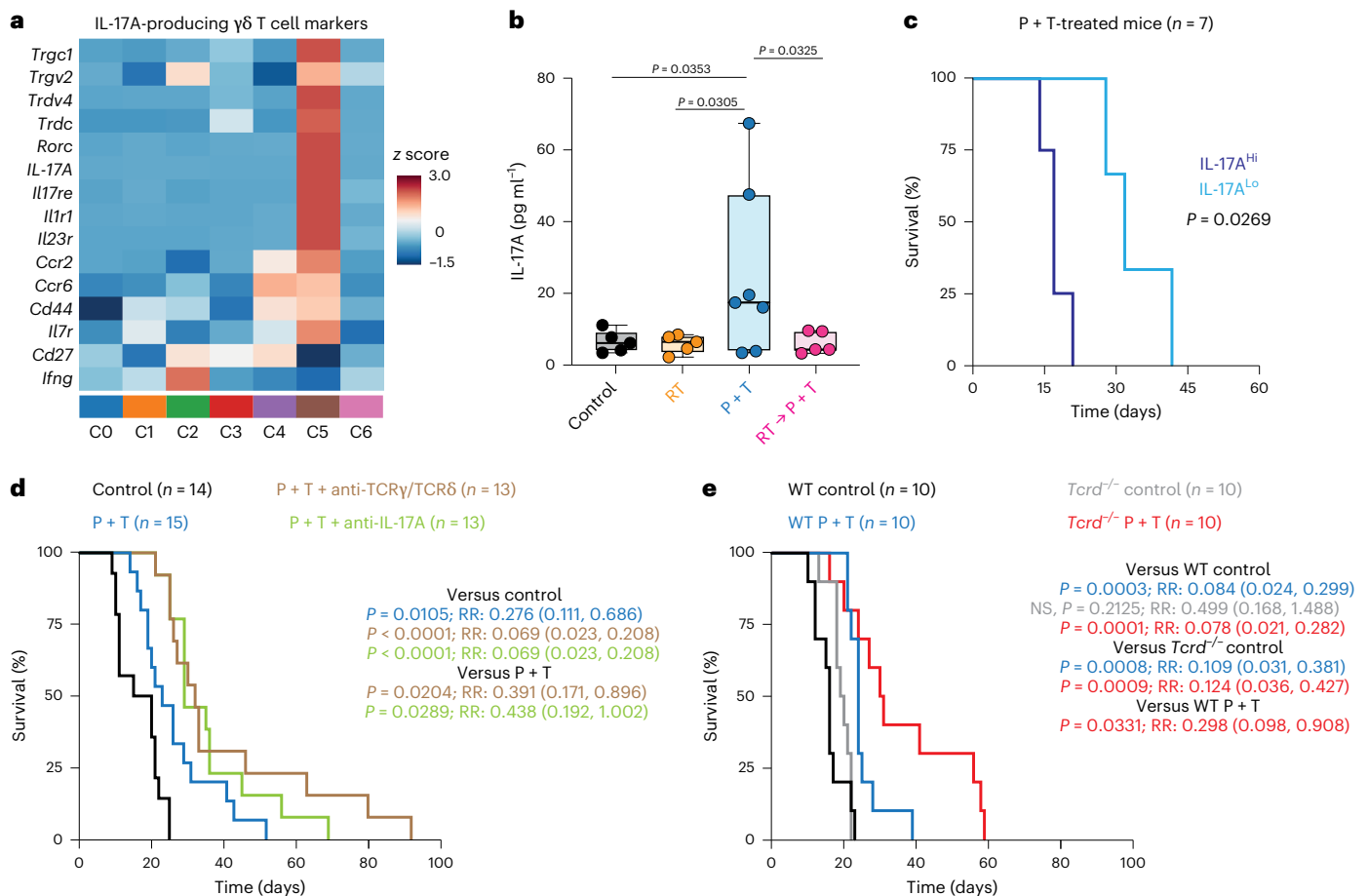


Fig. 2 | $\gamma\delta$ T cells elicited by CDK4/CDK6 inhibition mediate tumor-promoting effects. **a, Heat map of IL-17A-producing $\gamma\delta$ T cell-defining genes in each T cell subcluster. Data were obtained from 8,395 T cells (pooled from 27 tumors, with three independent biological samples per treatment group each obtained from individual mice) as reported in Fig. 1f,g. **b**, Serum levels of IL-17A in mice bearing M/D-driven tumors treated with RT (n = 5), P + T (n = 7) or RT → P + T (n = 5) or left untreated (control, n = 5) and assessed at day 6. Box plots show median $\pm$ interquartile range; whiskers indicate minima and maxima. Significant ($P < 0.05$) statistical differences are shown. **c**, Survival curves of mice bearing M/D-driven tumors treated with P + T following stratification based on median**

IL-17A serum levels as high ($\geq$ median; n = 4) and low ($<$ median; n = 3). **d**, Survival curves of mice bearing M/D-driven tumors left untreated or treated with P + T $\pm$ anti-TCR γ /TCR δ or anti-IL-17A. **e**, Survival curves of wild-type (WT) and *Tcrd*^{-/-} mice bearing M/D-driven tumors left untreated or treated with P + T. Number of mice, RR with 95% confidence interval and P values are shown in **d** and **e**. P values were determined by one-way ANOVA plus Fisher's LSD test (**b**), two-sided Gehan-Breslow-Wilcoxon test (**c** and **d**) and log-rank test (**e**). Part of the results shown in **d** are also depicted in Figs. 1b,e, 3f and 4i (control and P + T) and Extended Data Fig. 1b,f (control).

CD8⁺ proliferating T (T_{pr}) cells; Fig. 1g,h); (2) two subclusters expressing *Cd4* plus genes encoding T cell activation markers (*Cd40lg*, *Cd28*, *Il2r* and *Icos*; C1; CD4⁺ T_{eff} cells) or canonical T_{reg} cells markers (*Foxp3*, *Il2ra*, *Ilkzf2* and *Ctla4*; C4; T_{reg} cells); (3) one subcluster enriched in transcripts associated with natural killer T (NKT) cells (*Klrb1c*, *Klrd1*, *Sell*, *Satb1*, *Txk* and *Ly6c2*; C3; NKT cells) and (4) one subcluster characterized by the selective expression of *Trgc1*, *Trdc*, *Trdv4* and *Trgv2* (which all encode $\gamma\delta$ T cell antigen receptor (TCR) components) coupled to the lack of *Cd8b1*, *Cd8a* and *Cd4* expression (C5; $\gamma\delta$ T cells; Fig. 1g,h).

Focal RT alone caused a relative reduction in CD8⁺ T_{ext} cell infiltration ($P < 0.0001$, false discovery rate (FDR)-corrected full pairwise χ^2 test) plus a relative accumulation of CD8⁺ T_{eff} cells ($P < 0.0001$) and proliferating CD8⁺ T cells, as well as a minor increase in the relative abundance of $\gamma\delta$ T cells ($P < 0.0001$), but no alterations in T_{reg} cells (not significant, $P = 0.4991$) compared to untreated tumors (Fig. 1i). P + T-treated M/D-driven carcinomas, by contrast, were highly enriched in $\gamma\delta$ T cells ($P < 0.0001$) coupled with signs of increased CD8⁺ T_{eff} cell ($P < 0.0001$) and NKT cell ($P < 0.0001$) relative abundance, plus a partial reduction in the relative intratumoral levels of CD8⁺ T_{ext} ($P < 0.0001$) and T_{reg} cells ($P < 0.0001$; Fig. 1i). Finally, M/D-driven tumors exposed to RT → P + T not only showed minimal enrichment of $\gamma\delta$ T cells compared

to tumors receiving P + T alone ($P = 0.0001$) but also exhibited a smaller cluster of CD8⁺ T_{ext} cells than untreated tumors ($P < 0.0001$) or tumors receiving RT alone ($P < 0.0001$), coupled with higher relative abundance of CD4⁺ T_{eff} cells ($P = 0.0443$ versus untreated and $P = 0.0380$ versus P + T-treated tumors; Fig. 1i).

Importantly, flow cytometry demonstrated that the minor increase in relative $\gamma\delta$ T cell abundance detected by scRNA-seq in irradiated M/D-driven tumors (Fig. 1i) was not paralleled by an increase in the absolute number of intratumoral $\gamma\delta$ T cells (Fig. 1j), de facto reflecting the relative contraction of another CD3⁺ T cell compartment. Conversely, P + T elicited an absolute increase in intratumoral CD4⁺ CD8⁺ $\gamma\delta$ T cells that could be prevented when this therapeutic regimen was preceded by RT (Fig. 1j). Such an expansion exhibited some degree of oligoclonality (and hence was associated with a partial decrease in $\gamma\delta$ TCR diversity, at least with respect to δ TCR chains; Extended Data Fig. 1h).

Along with the superior efficacy of RT → P + T relative to P + T alone (Fig. 1a–c), these findings led us to hypothesize that a population of CD4⁺ CD8⁺ $\gamma\delta$ T cells may limit the therapeutic efficacy of CDK4/CDK6 inhibitors plus endocrine therapy because of local immunosuppression.

γδ T cells elicited by CDK4/CDK6 inhibition promote HR⁺HER2⁻ BC progression in mice

To obtain insights into the immunological profile of CD4⁻CD8⁻ γδ T cells infiltrating M/D-driven tumors, we initially focused on their transcriptional configuration. We found that, besides expressing *Trgc1*, *Trdc*, *Trdv4* and *Trgv2* (which all encode γδ TCR components), these cells are the sole tumor-infiltrating CD3⁺ cell population to encode the immunomodulatory cytokine IL-17A and IL-17 receptor E (IL-17RE; Fig. 2a and Extended Data Fig. 2a), which have previously been shown to promote disease progression in the triple-negative BC setting³². Moreover, CD4⁻CD8⁻ γδ T cells infiltrating M/D-driven tumors expressed the transcriptional factor RAR-related orphan receptor-γ (*Rorc*, best known as RORγ), the receptors for IL-7, IL-23 and IL-1β, chemokine (C-C motif) receptor 2 (*Ccr2*), *Ccr6* and *Cd44*, but not the co-stimulatory receptor *Cd27* or the effector molecule interferon-γ (*Ifng*; Fig. 2a and Extended Data Fig. 2a), delineating an overall immunosuppressive configuration associated with IL-17A secretion, as previously described in the setting of neoplastic lesions of the lung^{33–35}. Of note, a similar population of IL-17-producing V_γ2⁺V_δ4⁺ cells, although less frequent than its V_γ4⁺ counterpart, has previously been documented in mouse models of glioma^{36,37}.

Only mice with M/D-driven mammary tumors exposed to P + T exhibited a significant increase in circulating IL-17A over control levels (Fig. 2b), correlating with increased blood-borne concentrations of IL-1β and IL-23 (Extended Data Fig. 2b), for which CD4⁻CD8⁻ γδ T cells infiltrating M/D-driven tumors express cognate receptors (Fig. 2a and Extended Data Fig. 2a). Such a P + T-driven increase in circulating IL-17A was not observed when M/D-driven mammary carcinomas were established in C57BL/6 female mice lacking *Tcrd*³⁸ (Extended Data Fig. 2c), mechanistically implicating γδ T cells as a major source of IL-17A in this setting. Moreover, mice with M/D-driven tumors treated with P + T exhibiting higher-than-median circulating IL-17A 7 days after treatment initiation exhibited shortened OS than their counterparts with lower-than-median circulating IL-17A levels (Fig. 2c). To clarify the impact of CD4⁻CD8⁻ γδ T cells on the sensitivity of HR⁺HER2⁻ BC to CDK4/CDK6 inhibitors plus endocrine therapy, we exposed mice with M/D-driven tumors to P + T in combination with an antibody that antagonizes the γδ TCR or neutralizing anti-IL-17A. Both γδ TCR antagonism and IL-17A neutralization extended the OS benefit afforded by P + T to mice with M/D-driven carcinomas (Fig. 2d). Moreover, M/D-driven mammary carcinomas established in *Tcrd*^{-/-} mice were

more sensitive to P + T than their counterparts growing in wild-type mice (Fig. 2e).

Thus, interfering with IL-17A-producing CD4⁻CD8⁻ γδ T cells boosts the activity of CDK4/CDK6 inhibitors plus endocrine therapy in a pre-clinical model of HR⁺HER2⁻ BC.

CCL2 drives γδ T cell recruitment in response to CDK4/CDK6 inhibition

We next aimed to determine the mechanisms underlying the accumulation of IL-17A-producing CD4⁻CD8⁻ γδ T cells in M/D-driven carcinomas treated with P + T. As CDK4/CDK6 inhibitors are known to induce cellular senescence and abundant secretion of various chemokines⁹, we focused on genes encoding chemokine receptors overexpressed by γδ T cells compared to other CD3⁺ cells infiltrating M/D-driven tumors, *Ccr2* being one of the most differentially expressed (Fig. 2a and Extended Data Fig. 2a). We also observed that circulating IL-17A levels detected 7 days from treatment initiation exhibited a mild correlation with blood-borne CCL2 levels (Fig. 3a). Moreover, differential gene expression analysis (DGEA) of individuals with estrogen receptor⁺ (ER⁺) BC included in the public METABRIC dataset³⁹ revealed a twofold increase in *CCL2* levels following participant stratification based on a genetic signature for IL-17A production (Fig. 3b). Finally, CCL2 protein levels were increased in M/D-driven carcinomas exposed to P compared to their untreated counterparts (Fig. 3c).

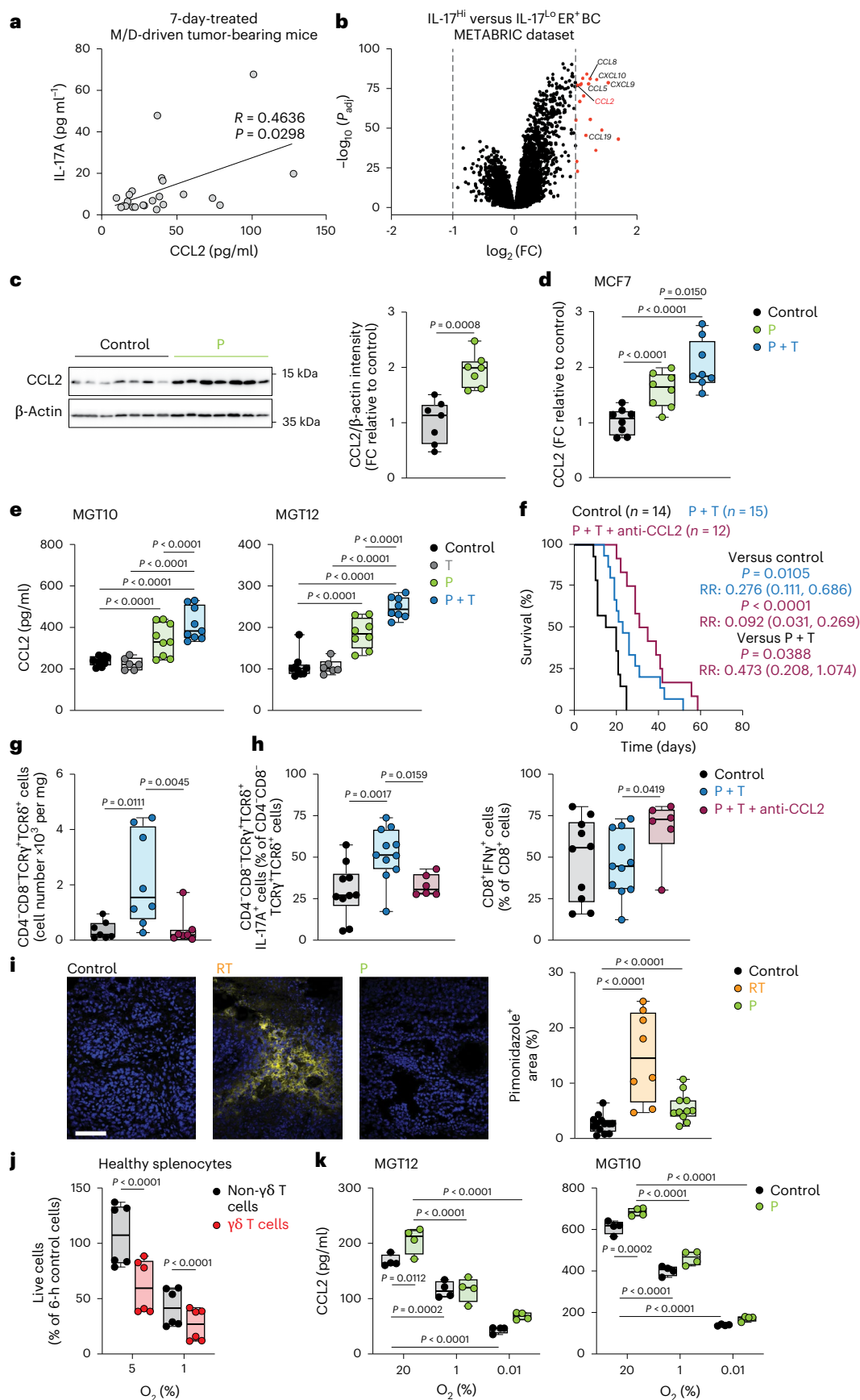
To formally assess the role of CCL2 in the accumulation of immunosuppressive CD4⁻CD8⁻ γδ T cells in M/D-driven tumors, we first tested whether P, T and their combination drive CCL2 secretion by cultured HR⁺HER2⁻ BC cells. We used a commercially available mouse HR⁺HER2⁻ BC cell line, namely, TS/A cells⁴⁰, two primary cell lines established in-house from untreated M/D-driven tumors (MGT10 and MGT12 cells)²¹ and commercially available human HR⁺HER2⁻ MCF7 cells⁴¹. In all models, T had no effect on CCL2 secretion, but P caused a significant increase in the amount of CCL2 detectable in the culture medium 72 h after treatment, an effect that, in the case of MGT10, MGT12 and MCF7 cells, was exacerbated by T (Fig. 3d,e and Extended Data Fig. 3a).

Consistent with these findings and the detrimental impact of IL-17A-producing CD4⁻CD8⁻ γδ T cells on the sensitivity of M/D-driven tumors to P + T, neutralization of circulating CCL2 with a specific antibody extended the OS benefit afforded by P + T (Fig. 3f), thus resembling γδ TCR antagonism and IL-17A neutralization (Fig. 2d). Further suggesting that CCL2 operates in the same immunological pathway,

Fig. 3 | CCL2 promotes γδ T cell recruitment following CDK4/CDK6 inhibition.

a, Correlation between circulating IL-17A and CCL2 levels in mice bearing M/D-driven tumors left untreated or treated as in Fig. 2b ($n = 22$ samples in total). P and R (Spearman's rank correlation coefficient) values are reported. **b**, Volcano plot of differentially expressed genes in individuals with ER⁺ BC in the METABRIC dataset ($n = 1,459$) stratified by median single-sample score for the [GO:0032740](https://doi.org/10.1038/s43018-025-01007-z) signature in high and low. Red dots indicate $\log_2$ (fold change) of >1 and adjusted P value (P_{adj}) of <0.0001 ; FC, fold change. **c**, CCL2 levels in M/D-driven tumors left untreated or treated with P (days 0–5) and collected at day 6 ($n = 7$ mice per group) expressed as fold change of CCL2 levels in untreated tumors after normalization to β-actin (from the same tumor and blots processed in parallel). **d,e**, CCL2 secretion from MCF7 ($n = 8$ independent biological samples from two independent experiments for each treatment; **d**), MGT10 ($n = 9$ independent biological samples from three independent experiments for control, P and P + T; $n = 6$ independent biological samples from two independent experiments for T) and MGT12 ($n = 9$ independent biological samples from three independent experiments for control; $n = 8$ independent biological samples from three independent experiments for P and P + T; $n = 6$ independent biological samples from two independent experiments for T; **e**) cells exposed to the indicated treatments for 72 h. **f**, Survival curves of mice with M/D-driven tumors randomly allocated to receive no treatment or P + T ± anti-CCL2. The number of mice, RR with 95% confidence interval and P values are shown. **g**, Density of CD4⁻CD8⁻ TCRγ⁺TCRδ⁺ (γδ T) cells infiltrating M/D-driven tumors treated with P + T ($n = 8$) or P + T + anti-CCL2 ($n = 7$) or left untreated ($n = 7$) and collected

at the experimental endpoint. **h**, Frequency of CD4⁻CD8⁻ TCRγ⁺TCRδ⁺IL-17A⁺ and CD8⁺IFNγ⁺ cells infiltrating M/D-driven tumors treated with P + T ($n = 11$) or P + T + anti-CCL2 ($n = 6$) or left untreated ($n = 10$) after ex vivo stimulation with PMA/ionomycin. **i**, Representative images and quantification of pimonidazole⁺ areas in M/D-driven tumors treated with RT (days 0–2; $n = 8$ images from two tumors) or P (days 0–5; $n = 11$ images from two tumors) or left untreated ($n = 13$ images from three tumors) and collected at day 6; scale bar, 50 μm. **j**, Frequency of live non-γδ T and γδ T cells after 72 h of stimulation of healthy splenocytes with anti-CD3 and IL-2 under hypoxic conditions ($n = 6$ independent biological samples per group pooled from two independent experiments). **k**, CCL2 secretion from MGT10 and MGT12 cells exposed to P under normoxic or hypoxic conditions for 72 h ($n = 4$ independent biological samples per group). In **c–e** and **g–k**, box plots show median ± interquartile range, and whiskers indicate minima and maxima. Relevant and significant ($P < 0.05$) statistical differences are shown. P values were determined by two-sided Spearman's rank correlation test (**a**), the limma method based on moderated t -statistics (**b**), two-tailed unpaired t -test (**c**), linear mixed effects regression followed by simultaneous tests of general linear hypotheses (**d**, **e** and **i–k**), two-sided Gehan–Breslow–Wilcoxon test (**f**), one-way ANOVA plus Fisher's LSD test (**g** and **h**, left) and Kruskal–Wallis test with uncorrected Dunn's test (**g** and **h**, right). Part of the results shown in **f** are also depicted in Figs. 1b,e, 2d and 4i (control and P + T) and Extended Data Fig. 1b,f (control), and part of the results shown in **g** (control and P + T) are also depicted in Fig. 1j.



the TME of M/D-driven tumors relapsing on P + T in the context of CCL2 neutralization was depleted of CD4⁺CD8⁺ $\gamma\delta$ T cells (Fig. 3g) and more specifically of IL-17A⁺ $\gamma\delta$ T cells (Fig. 3h), coupled with an increase in tumor-infiltrating CD8⁺ T cells expressing IFN γ following ex vivo stimulation (Fig. 3h).

We next investigated the mechanisms through which RT prevents $\gamma\delta$ T cell recruitment to M/D-driven tumors subsequently exposed to P + T. We initially thought that $\gamma\delta$ T cells could be more radiosensitive than their non- $\gamma\delta$ CD3⁺ T cell counterparts. However, such a differential radiosensitivity would not explain our in vivo observations because we invariably delivered RT to M/D-driven tumors before P + T, hence irradiating a TME virtually devoid of $\gamma\delta$ T cells (Fig. 1i,j). We therefore hypothesized that RT alters the M/D-driven TME to render it incompatible with the recruitment or survival of $\gamma\delta$ T cells, initially focusing on hypoxia. As part of conventional radiation responses, non-ablative RT (at least initially) selects for hypoxic tumor areas, owing to their intrinsic radioresistance²². In line with this notion, the TME of irradiated M/D-driven carcinomas stained positively for the hypoxia-sensitive probe pimonidazole (Fig. 3i and Extended Data Fig. 3b). Moreover, although conventional $\alpha\beta$ T cells activated ex vivo by a CD3 agonist plus IL-2 tolerated oxygen levels as low as 1%, the same was not true for $\gamma\delta$ T cells, which manifested proliferative defects coupled to cell death starting at oxygen levels of 5% (and culminating at 1%; Fig. 3j). Moreover, MGT10 and MGT12 cells cultured under hypoxic conditions exhibited considerable defects in CCL2 secretion, both under untreated conditions and following treatment with P (Fig. 3k).

These data delineate a detrimental, CCL2-driven mechanism through which HR⁺HER2⁺ BC cells exposed to CDK4/CDK6 inhibitors recruit hypoxia-sensitive, IL-17A-producing CD4⁺CD8⁺ $\gamma\delta$ T cells that are associated with treatment resistance.

CX3CR1⁺ TAMs assist $\gamma\delta$ T cells to drive HR⁺HER2⁺ BC progression in mice

Comparing the myeloid compartment of P + T-treated versus untreated M/D-driven tumors, we noticed that $\gamma\delta$ T cell infiltration as promoted by P + T is accompanied by a global reprogramming of myeloid cells toward an immunosuppressive phenotype characterized by the upregulation of genes encoding TAM-related (*Apoe*, *Mrc1* and *Cd63*) and complement-related (*C1qa*, *C1qb* and *C1qc*) factors, coupled with the downregulation of genes involved in type 1 IFN signaling (*Stat1*, *Stat3*, *Cxcl9*, *Cxcl10*, *Irf5* and *Irf7*) and antigen presentation (*B2m*, *H2-Aa*, *H2-Ab1*, *H2-D1*, *H2-Eb1*, *H2-K1*, *H2-Q4*, *H2-Q6*, *H2-Q7*, *H2-T22* and *H2-T23*), which together define a core TAM signature in different human cancers (including BC^{42,43}; Extended Data Fig. 4a).

Reclustering the 31,723 myeloid cells (Extended Data Fig. 4b,c) that we profiled by scRNA-seq identified ten subclusters, with the majority (C4–C9) displaying signatures of TAMs, a minority showing expression of DC-related genes (C1–C3) and one subcluster displaying the monocyte markers *S100a8*, *S100a9*, *Cxcl2* and *Cxcr2* (C0; Fig. 4a,b). The three DC subclusters were characterized by the selective expression of *Batf3* (C1; cross-presenting DCs), major histocompatibility complex (MHC) class II molecules plus *S100a4*, *Cd209a* and *Itgae* (C2; activated DCs) or signatures of plasmacytoid DCs, including *Ly6c2*, *Siglech*, *Cd7* and *Bcl11a* (C3; plasmacytoid DCs; Fig. 4a,b). Three CD11b⁺ TAM subclusters were characterized by the expression of proangiogenic and immunosuppressive factors such as *Vegfa*, *Fcgr2b*, *Trem2* and *Arg1* (C4; angiogenic TAMs), markers of M2-like polarization such as *Mrc1*, *Folr2* and high *Apoe* levels (C5; M2-like TAMs) or activation/proliferation markers including *Ly6c2*, *Ccr2* and *Itgam* (C6; CD11b⁺Ly6c⁺ TAMs; Fig. 4a,b). Other CD11c⁺ TAM subclusters were characterized by the common expression of MHC class II molecules plus *Mki67* (C7; proliferative CD11c⁺ TAMs), high *Apoe* levels, *C1qa* and *Cx3cr1* (C8; CX3CR1⁺ TAMs) or *Ly6a* (also known as *Sca1*; C9; SCA1⁺ TAMs; Fig. 4a,b). Importantly, P + T-driven $\gamma\delta$ T cell infiltration was closely accompanied by a relative enrichment in CX3CR1⁺ TAMs ($P < 0.0001$ versus untreated

tumors; FDR-corrected full pairwise χ^2 test), an effect that was reduced by RT ($P < 0.0001$ versus P + T-treated tumors; Fig. 4c,d).

DGEA revealed that, compared to their CX3CR1^{Lo} counterparts, CX3CR1^{Hi} cells express increased levels of immunosuppressive genes and markers of M2-like TAMs, including *Apoe*, *Mrc1*, *Adgre1*, *Cd63*, *Arg1*, *Il10*, *Mmp12* and *Mmp14*, and complement-encoding genes (*C1qa*, *C1qb* and *C1qc*) but reduced levels of genes involved in type 1 IFN signaling (*Stat1*, *Stat3*, *Cxcl9*, *Cxcl10*, *Irf1*, *Irf5* and *Irf7*) and antigen presentation (*H2-DMb2*, *H2-Eb2*, *H2-Oa*, *H2-Ob*, *H2-Q4*, *H2-Q6*, *H2-T22*, *H2-T23* and *H2-T24*; Fig. 4e,f), suggesting an immunosuppressive and tumor-promoting status⁴⁴. P + T further shifted the profile of CX3CR1^{Hi} TAMs toward an even more immunosuppressive configuration characterized by further upregulation of *Mrc1* and *Mmp12*, coupled to the further downregulation of genes involved in antigen presentation (*B2m*, *H2-Aa*, *H2-Ab1*, *H2-D1*, *H2-Eb1*, *H2-K1*, *H2-Q4*, *H2-Q6*, *H2-Q7*, *H2-T22* and *H2-T23*) and type 1 IFN signaling (*Stat1*, *Stat3*, *Cxcl9*, *Irf1* and *Irf5*; Fig. 4e,f), which is in line with global transcriptomic changes exhibited by myeloid cells from P + T-treated versus untreated M/D-driven tumors (Extended Data Fig. 4a).

CX3CR1^{Hi} TAMs from the M/D-driven TME expressed increased levels of IL-17A receptor (Fig. 4g) and exhibited an increased genetic signature of IL-17 responsiveness compared to CX3CR1^{Lo} TAMs (Fig. 4h). Further suggesting a mechanistic link between $\gamma\delta$ T cells and immunosuppressive CX3CR1^{Hi} TAMs, the ability of P + T to cause the accumulation of CX3CR1^{Hi} TAMs was largely compromised by the coadministration of an antibody targeting $\gamma\delta$ TCR as well as in mice lacking *Tcrd* (Extended Data Fig. 4d). Similarly, administration of neutralizing anti-IL-17 not only ameliorated the efficacy of P + T (Fig. 2d) but also limited intratumoral CX3CR1⁺ TAM levels (Extended Data Fig. 4d). Moreover, myeloid cells isolated from M/D-driven tumors exposed to P + T plus an IL-17 neutralizer (but not P + T alone) expressed activation markers such as CD80 and MHC class II molecules in response to immunostimulatory cues (that is, lipopolysaccharide and IFN γ ; Extended Data Fig. 4e). Corroborating the detrimental role for CX3CR1^{Hi} TAMs downstream of $\gamma\delta$ T cell infiltration, repolarization of the myeloid TME with an antibody targeting colony stimulating factor 1 receptor (CSF1R) ameliorated the therapeutic efficacy of P + T against M/D-driven tumors (Fig. 4i), an effect that could not be further extended by concomitant IL-17 neutralization (Fig. 4i).

These findings suggest that CX3CR1^{Hi} TAMs operate downstream of $\gamma\delta$ T cells and IL-17 to establish immunosuppression in the TME of M/D-driven carcinomas responding to P + T.

Impact of $\gamma\delta$ T cells, CCL2 and CX3CR1⁺ TAMs on human HR⁺HER2⁺ BC

To investigate the translational relevance of our findings, we took advantage of the METABRIC dataset³⁹ plus five distinct cohorts of individuals with HR⁺ BC for whom biopsies or blood samples were available for analysis.

Analysis of the METABRIC database revealed that individuals with ER⁺ (but not ER[−]) BC with genetic signatures for $\gamma\delta$ T cell infiltration ($\gamma\delta^{\text{Hi}}$) or positive regulation of IL-17 production (IL-17^{Hi}) in the TME experienced inferior disease-specific survival (Fig. 5a and Extended Data Fig. 5a) and OS (Extended Data Fig. 5b,c) compared to their low ($\gamma\delta^{\text{Lo}}$ or IL-17^{Lo}, respectively) counterparts. Moreover, grade 3 ER⁺ tumors from the METABRIC database exhibited higher signatures for $\gamma\delta$ T cell infiltration or positive regulation of IL-17 production than their grade 1 and grade 2 counterparts (Fig. 5b).

Corroborating a detrimental effect of $\gamma\delta$ T cells, tumor infiltration by this immune cell population as assessed in diagnostic biopsies was considerably higher for grade 3 tumors than for grade 1 or 2 tumors in a cohort of 84 participants with HR⁺HER2⁺ BC (Fig. 5c). Similar findings were obtained for CD8⁺ T cells (Extended Data Fig. 5d). However, although $\gamma\delta$ T cells detected in grade 3 HR⁺HER2⁺ BC samples were mostly intratumoral, CD8⁺ T cells mainly resided in the tumor stroma

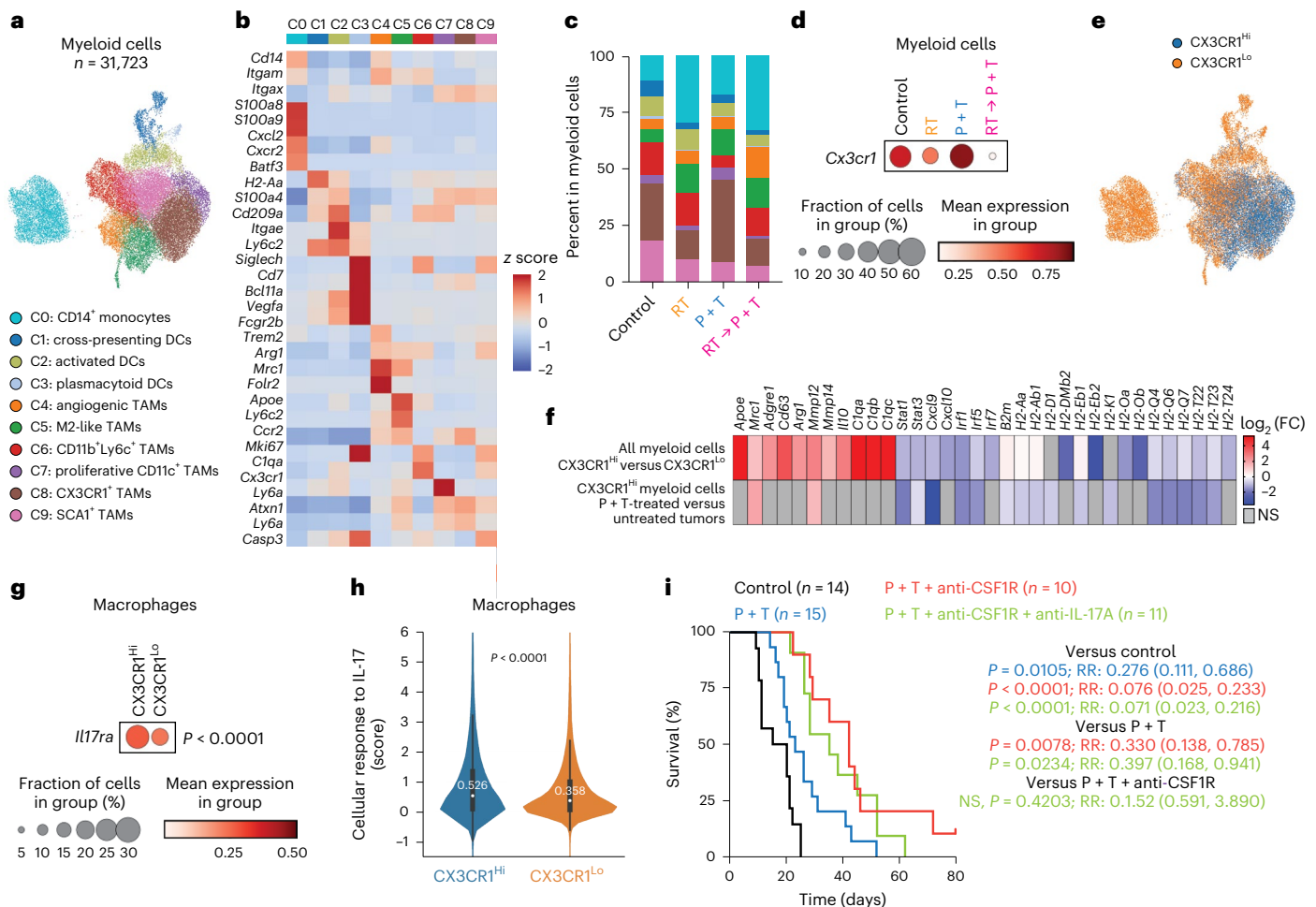


Fig. 4 | Resistance to CDK4/CDK6 inhibitors is linked to the accumulation of CX3CR1⁺ macrophages in M/D-driven mammary carcinomas. **a**, UMAP subclustering of myeloid cells in ten phenotypes, as indicated by the color-coded legend. Data are pooled from 27 independent tumor samples as indicated in Fig. 1f. **b**, Heat map of cluster-defining genes for the ten myeloid cell phenotypes. **c**, Bar plots showing the frequency (average of three independent biological samples per treatment group each obtained from individual mice) of each myeloid cell subcluster in the indicated treatment groups. **d**, Expression of *Cx3cr1* in myeloid cells from the indicated treatment groups (average of three independent biological samples per treatment group each obtained from individual mice). **e**, UMAP of CX3CR1^{hi} and CX3CR1^{lo} cells by scRNA-seq. **f**, Heat map of TAM-defining genes in CX3CR1^{hi} versus CX3CR1^{lo} myeloid cells and in CX3CR1^{hi} cells from P + T-treated M/D-driven tumors versus

untreated M/D-driven tumors. **g**, Dot plot of *Il17ra* expression in CX3CR1^{hi} and CX3CR1^{lo} macrophages. **h**, Violin plots depicting the score distribution for the GO:0097398 signature in CX3CR1^{hi} and CX3CR1^{lo} macrophages. Within each violin plot, box plots report median $\pm$ interquartile range (whiskers indicate minima and maxima). **i**, Survival curves of mice bearing M/D-driven tumors randomly allocated to receive no treatment, P + T $\pm$ anti-CSF1R alone or combined with anti-IL-17A. Number of mice, RR with 95% confidence interval and *P* values are shown. *P* values were determined by two-sided Mann–Whitney–Wilcoxon test (**g** and **h**) and two-sided Gehan–Breslow–Wilcoxon test (**i**). For data shown in **c**, FDR-corrected *P* values from a full pairwise χ^2 statistical test between treatments for each cluster are reported in the main text. Part of the results shown in **i** are also depicted in Figs. 1b, e, 2d and 3f (control and P + T) and Extended Data Fig. 1b, f (control).

(Fig. 5c and Extended Data Fig. 5d). Intriguingly, although the overall abundance of CD68⁺ macrophages was not influenced by tumor grade, grade 3 HR⁺HER2⁺ BC samples had lower intratumoral and higher stromal CD68⁺ macrophages than their grade 1 and 2 counterparts (Extended Data Fig. 5e).

Intrigued by these findings, we performed multispectral immunofluorescence on diagnostic biopsies from individuals with HR⁺HER2⁺ BC ($n = 26$), including antibodies specific to the $\gamma\delta$ TCR, CD68 and CD274 (best known as PD-L1), which is critical for tumor-driven immunosuppression in multiple cancer types⁴⁵. This demonstrated that $\gamma\delta$ T cells reside significantly closer to PD-L1⁺ macrophages and malignant cells than their PD-L1[−] counterparts (Extended Data Fig. 6a), further suggesting a role for these lymphoid cells in the establishment of immunosuppression in the TME.

Next, we performed scRNA-seq analysis of CD45⁺ cells isolated from longitudinal biopsies of 12 individuals with high-risk, early-stage

HR⁺HER2[−] BC enrolled in a clinical trial of preoperative PD-1 inhibition with pembrolizumab, followed by pembrolizumab plus focal RT to the primary breast lesion in three fractions of 8 Gy each (NCT03366844)⁴⁶. More specifically, we had access to 12 baseline samples, 12 samples after PD-1 blockade and 12 surgical samples, overall demonstrating that human $\gamma\delta$ T cells infiltrating the HR⁺HER2[−] BC microenvironment also exhibit the downregulation of transcriptional signatures linked to immunostimulation, including (but not limited to) signatures of IFN γ signaling and other effector functions (Fig. 5d). Although they did not form a distinct population based on global clustering, compared to their CX3CR1[−] counterparts, CX3CR1⁺ TAMs from this dataset exhibited the downregulation of transcriptional programs linked to efficient anticancer immunity, encompassing signatures of T cell activation, inflammatory responses and type I IFN signaling (Fig. 5d) as well as an enrichment of signatures linked to IL-17 signaling (Fig. 5e). Moreover, individuals with HR⁺HER2[−] BC that achieved a pathological complete

response at surgery ($n = 8$) exhibited limited amounts of CX3CR1⁺ macrophages compared to nonresponders ($n = 4$), both at baseline and at surgery ($P < 0.0001$, two-sided Mann–Whitney–Wilcoxon test; Fig. 5f and Extended Data Fig. 6b).

Next, we used the first set of blood samples from 28 individuals with advanced/metastatic HR⁺HER2⁻ BC collected before treatment with CDK4/CDK6 inhibitors plus endocrine therapy (baseline), at the end of the third treatment cycle (re-evaluation) and at disease relapse (for progressing individuals) to assess whether the number of circulating $\gamma\delta$ T cells would correlate with progression-free survival (PFS). We found that a higher-than-median circulating abundance of IL-7R⁺ (activated) CD3⁺CD4⁺CD8⁻ $\gamma\delta$ T cells at baseline (available for a subset of 23 participants) was associated with shortened PFS (Fig. 5g). These observations were corroborated by univariate Cox regression analysis ($P = 0.006$).

We also had access to existing data on circulating immunophenotype at baseline from a second cohort of 29 participants with advanced/metastatic HR⁺HER2⁻ BC who were similarly treated with CDK4/CDK6 inhibitors plus endocrine therapy. Importantly though, the circulating immunophenotype of these 29 participants had previously been determined for an independent project (not specifically assessing $\gamma\delta$ T cells) and hence could not be directly interrogated for $\gamma\delta$ T cell abundance. With that said, merging existing baseline data on circulating CD3⁺CD4⁺CD8⁻ T cells (most of which are $\gamma\delta$ T cells)^{47,48} from this latter cohort ($n = 29$) with baseline data on circulating CD3⁺CD4⁺CD8⁻ T cells as obtained from a subset ($n = 23$) of participants from the first cohort lent extra support (although with modest statistical significance) to the negative association between circulating $\gamma\delta$ T cells at baseline and PFS in participants with advanced/metastatic HR⁺HER2⁻ BC receiving CDK4/CDK6 inhibitors plus endocrine therapy (Extended Data Fig. 7a). Univariate Cox regression analysis confirmed the validity of these observations, although with modest significance ($P = 0.042$).

Of note, nonprogressing individuals from the former cohort of 28 participants exhibited reduced levels of blood-borne IL-7R⁺CD4⁺CD8⁻ $\gamma\delta$ T cells at baseline and at re-evaluation compared to participants who progressed (evaluable in $n = 8$ nonprogressing and $n = 16$ progressing individuals; Fig. 5h), in the context of comparable median follow-up (21.57 months for nonprogressing and 17.31 months for progressing individuals, $P = 0.2246$, unpaired t -test). Finally, an increase in circulating CCL2 could be detected as early as at re-evaluation in 16 of 28 participants from this cohort who would later progress (but not in 12 of 28 nonprogressing participants), and such an increase persisted at disease relapse (Fig. 5h). Intriguingly, the circulating levels of active CD4⁺CD8⁻ $\gamma\delta$ T cells also exhibited a mild positive correlation with the

circulating abundance of exhausted-like PD-1⁺CD8⁺ T cells and T_{reg} cells (Extended Data Fig. 7b).

Finally, we performed multispectral immunofluorescence on paired biopsies from eight individuals with advanced/metastatic HR⁺HER2⁻ BC collected at baseline and after disease progression on CDK4/CDK6 inhibitors plus endocrine therapy. In this setting, we were unable to detect any longitudinal alterations in the abundance of tumor-infiltrating PD1⁺CD8⁺ T cells, CD4⁺ T cells or FOXP3⁺ T_{reg} cells at relapse (Fig. 5i and Extended Data Fig. 7c,d). Conversely, tumor-infiltrating $\gamma\delta$ T cells were significantly increased in post- versus pretreatment biopsies (Fig. 5i).

Collectively, our findings suggest that a CCL2-driven mechanism of resistance to CDK4/CDK6 inhibitors mediated by IL-17-producing $\gamma\delta$ T cells and potentially executed by CX3CR1⁺ TAMs may be operational in individuals with HR⁺HER2⁻ BC.

Discussion

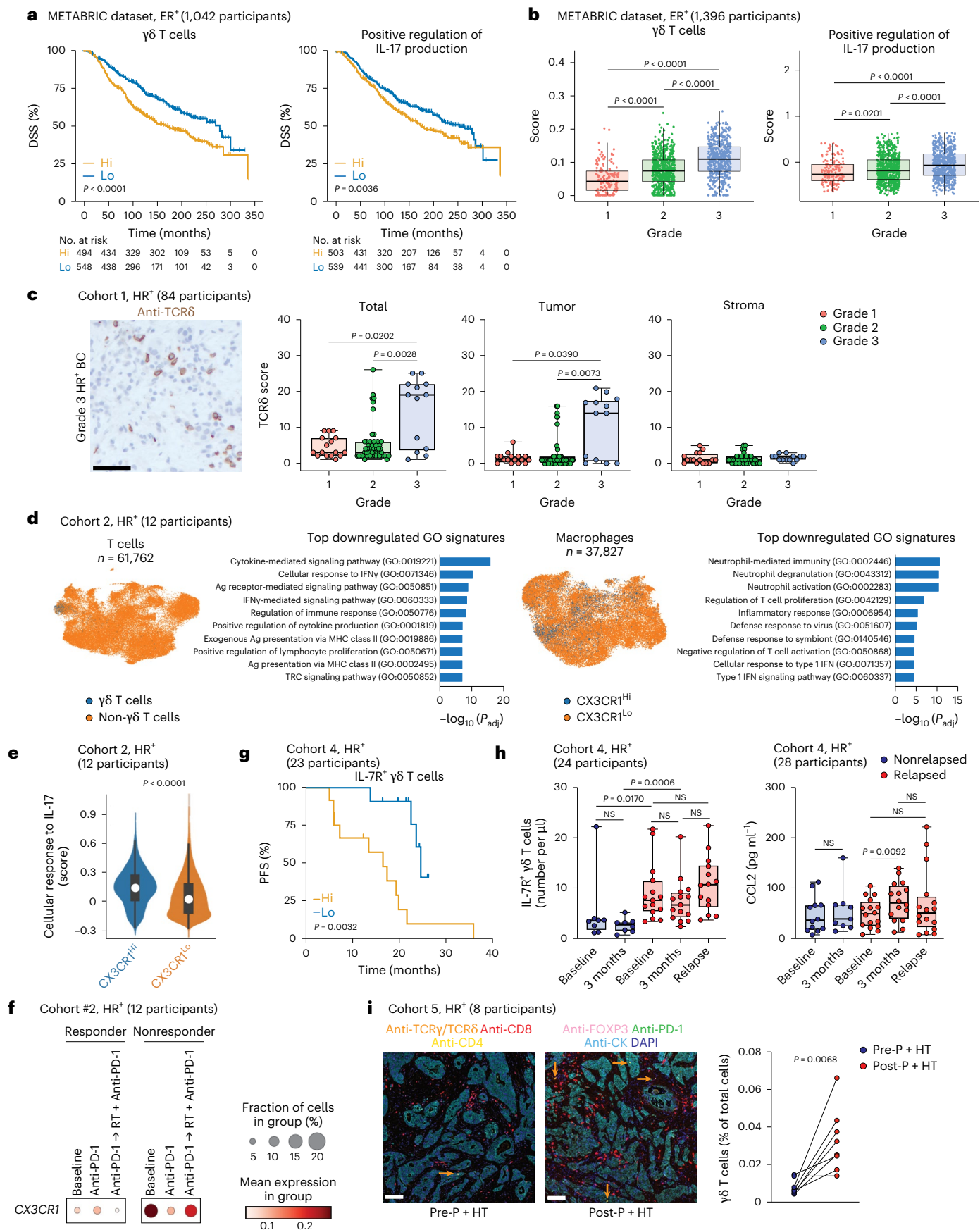
Here, we unveiled the ability of CDK4/CDK6 inhibitors to drive CCL2 secretion by HR⁺ BC cells, resulting in the recruitment of IL-17A-producing $\gamma\delta$ T cells to the TME coupled to the repolarization of TAMs toward an immunosuppressive CX3CR1⁺ phenotype that supports treatment resistance. Importantly, such a detrimental effect can be averted by focal RT delivered before treatment with CDK4/CDK6 inhibitors, at least partially owing to suppressed CCL2 secretion by hypoxic tumor areas that are (at least initially) selected by RT owing to their intrinsic radioresistance.

Our findings mostly originate from a single model of HR⁺ mammary carcinogenesis, namely M/D-driven tumors developing in immunocompetent female C57BL/6 mice²¹. Although this represents a caveat of our studies, it largely reflects our focus on a clinically relevant scenario (HR⁺HER2⁻ BC is the sole indication for which CDK4/CDK6 inhibitors are approved for use in humans) coupled to the following features and limitations of other common models of BC. First, the need for syngeneic hosts intrinsically linked to the study of cancer immunology in vivo restricts the choice of experimental model to mouse systems. Second, commonly available mouse BC cell lines that are truly HR⁺ and HER2⁻ such as TS/A and 67NR cells^{40,49} (1) do not recapitulate the clinical heterogeneity of BC, as they progress in a virtually identical manner in all mice, (2) fail to mimic early immunoediting and immuno-evasion as they have escaped immune control in their primordial host and are ultimately used (following considerable selection in vitro) to generate neoplasms in otherwise tumor-naïve and immunologically untrained hosts²⁶ and, most importantly, (3) are unsuitable to study CDK4/CDK6 inhibitors in vivo as they are intrinsically insensitive to

Fig. 5 | Impact of $\gamma\delta$ T cells and CX3CR1⁺ TAMs on human HR⁺ BC.

a, Disease-specific survival (DSS) curves for individuals with ER⁺ BC from the METABRIC database stratified by median single-sample $\gamma\delta$ T cell or IL-17 signature. **b**, Score for $\gamma\delta$ T cell and IL-17 signatures in ER⁺ breast tumors of different grades (grade 1: $n = 162$; grade 2: $n = 683$; grade 3: $n = 551$) from METABRIC. **c**, Representative image (left) and quantification (right) of TCR δ expression in diagnostic HR⁺HER2⁻ BC biopsies of the indicated grade (grade 1: $n = 16$; grade 2: $n = 55$; grade 3: $n = 13$). **d**, UMAPs for T cell and macrophage clusters by scRNA-seq color-coded for the indicated cell type and gene set enrichment analysis (GSEA) of downregulated genes in $\gamma\delta$ T versus non- $\gamma\delta$ T cells and CX3CR1^{hi} versus CX3CR1^{lo} macrophages from a total of 36 longitudinal breast biopsies obtained from 12 individuals with HR⁺HER2⁻ BC at three time points (at baseline, after PD-1 treatment and at surgery). Significance of the indicated top ten downregulated Gene Ontology (GO) signatures is reported; Ag, antigen. **e**, Violin plots depicting the score of the GO:0097398 signature in CX3CR1^{hi} and CX3CR1^{lo} macrophages. Within each violin plot, box plots report median $\pm$ interquartile range (whiskers indicate minima and maxima). **f**, Average expression of CX3CR1 in myeloid cells from the indicated treatment groups among responders ($n = 8$) and nonresponders ($n = 4$). **g**, PFS curves for individuals with HR⁺ BC stratified based on the median abundance of circulating IL-7R⁺ $\gamma\delta$ T cells at baseline in high ($\geq$ median; $n = 12$) and low ($<$ median; $n = 11$).

h, Circulating IL-7R⁺ $\gamma\delta$ T cells (left) and CCL2 levels (right) in longitudinal blood samples collected at baseline, at re-evaluation (after 3 months) and at disease progression from individuals with HR⁺ BC on CDK4/CDK6 inhibitors plus hormone therapy (HT). Data from participants who did not relapse ($n = 8$ for $\gamma\delta$ T cells at baseline and 3 months; $n = 12$ for CCL2 at baseline; $n = 9$ for CCL2 at 3 months) and those that did relapse ($n = 15$ for $\gamma\delta$ T cells at baseline and 3 months; $n = 14$ for $\gamma\delta$ T cells at relapse; $n = 16$ for CCL2 at all time points) are shown. **i**, Representative images and quantification of $\gamma\delta$ T cells (indicated by arrows) of paired HR⁺ tumor biopsies collected at baseline and after disease progression on P + HT. The numbers of participants for each cohort are indicated. In **b**, **c** and **h**, box plots overlaid by individual data points show median $\pm$ interquartile range; whiskers indicate minima and maxima. Relevant and significant ($P < 0.05$) statistical differences are shown. Relevant lack of statistical significance is also indicated in **h**. P values were determined by log-rank test (**a** and **g**), Mann–Whitney–Wilcoxon two-sided test (**b** and **e**), Kruskal–Wallis test with an uncorrected Dunn's test (**c**), two-sided Wilcoxon matched-pairs (for differences between longitudinal samples from the same participant) or unpaired (for differences between participants who had relapsed and those who did not relapse) signed rank test (**h**) and two-tailed paired t -test (**i**). For data shown in **f**, P values from Mann–Whitney–Wilcoxon two-sided tests between relevant groups are reported in the main text; scale bars, 50 μ m (**c**) and 100 μ m (**i**).



them¹⁶. Finally, the sole genetically engineered mouse model that expresses HRs (namely, MMTV-PyMT mice) (1) highly expresses HER2 (refs. 50,51) and (2) loses HR expression by the time mice are randomized to treatment^{9,50,51} or tumors are used to generate cell lines^{9,50,51}, de facto modeling another type (HER2⁺) of BC. Unfortunately, to the best of our knowledge, no other syngeneic model of bona fide HR⁺HER2⁻ BC that is insensitive to PD-1 blockade and responsive to CDK4/CDK6 inhibitors plus endocrine therapy, de facto recapitulating key therapeutic aspects of the human disease^{27,52–54}, is available to extend our observations in a clinically relevant manner.

Of note, we demonstrate that CD4⁺ and CD8⁺ T cell depletion fails to impair the systemic efficacy of P + T in a bona fide model of HR⁺ BC, contrary to previous data from melanoma^{30,31} and HER2⁺ BC⁹. Besides its direct clinical relevance, this finding reaffirms the fundamental difference between the immunobiology of distinct tumor types, including cancers originating from the same tissue. Conversely, CD4⁺ and CD8⁺ T cell depletion limited the efficacy of RT → P + T in our model, which is in line with the established ability of RT to elicit therapeutic responses that (at least in some settings) involve adaptive immunity⁵⁵.

Data from multiple clinical cohorts suggest that the pathway we identified may also be operational in individuals with HR⁺ BC, potentially delineating an actionable axis to increase the efficacy of CDK4/CDK6 inhibitors in the clinic. This may be especially important for individuals with HR⁺ BC who do not respond to first-line CDK4/CDK6 inhibitors, as these individuals experience very poor survival⁵⁶. Indeed, current and previous data from our group demonstrate that RT can be harnessed to prevent the infiltration of M/D-driven carcinomas by $\gamma\delta$ T cells, at least partially owing to the RT-driven selection of hypoxic and thus radioresistant TME regions and hence to extend the therapeutic benefits of subsequently (but not concomitantly or antecedently)¹⁶ delivered P + T. Thus, RT stands out as a promising tool to improve the efficacy of CDK4/CDK6 inhibitors plus endocrine therapy in individuals with advanced/metastatic HR⁺HER2⁻ BC, and a randomized phase 2 clinical trial is ongoing to formally assess this possibility in individuals with oligometastatic HR⁺ BC (less than five metastases, no cranial involvement; CIMER, [NCT04563507](https://clinicaltrials.gov/ct2/show/study?term=NCT04563507)). Our current findings also suggest that IL-17 blockers (three of which are currently approved for the treatment of psoriasis and other autoimmune conditions, for example, secukinumab, ixekizumab and brodalumab)⁵⁷ may constitute promising combinatorial partners for CDK4/CDK6 inhibitor treatment plus endocrine therapy for individuals with advanced HR⁺HER2⁻ BC who cannot receive RT. Exploratory clinical trials are needed to assess the safety and therapeutic potential of such a combinatorial regimen, but an attentive evaluation of potential adverse events is necessary, as some clinical trials testing CDK4/CDK6 inhibitors plus endocrine therapy and PD-1 blockade were discontinued due to safety signals⁵⁸.

The role of $\gamma\delta$ T cells in cancer remains debated. Indeed, although in some settings, including glioblastoma⁵⁹, mismatch repair-deficient colorectal cancer lacking MHC class I expression⁶⁰ and triple-negative BC^{61–63}, $\gamma\delta$ T cells appear to exert antitumor activity, in others, such as pancreatic cancer, $\gamma\delta$ T cells have been shown to mediate robust immunosuppressive effects⁶⁴. At least in part, this reflects the existence of two distinct polarization fates that are dictated during thymic $\gamma\delta$ T cell development⁶⁵: a type 1 helper T cell-like IL-17⁺CD27⁺ IFN γ -secreting state with potent antitumor effects⁶⁰ versus an IL-17-producing helper T cell-like CD27⁺IFN γ ⁻IL-17-secreting state with tumor-supporting functions³². $\gamma\delta$ T cells bearing specific TCR configurations have been associated with pronounced immunosuppressive (for example, V γ 6⁺V δ 1⁺ cells)³⁵ versus immunostimulatory (for example, V γ 9⁺V δ 2⁺ cells) effects⁶⁶, but the precise impact of $\gamma\delta$ TCR chains on functional $\gamma\delta$ T cell polarization has yet to be elucidated. In our setting, IL-17-producing $\gamma\delta$ T cells that infiltrate M/D-driven mammary carcinomas in response to P + T appear to express V γ 2 and V δ 4, a relatively uncommon $\gamma\delta$ TCR configuration that has previously been documented in mouse models of glioma^{36,37}. With that said, although determining the clonality of the

$\gamma\delta$ T cells that infiltrate HR⁺ BCs in response to CDK4/CDK6 inhibitors is of interest, $\gamma\delta$ T cell polarization and activation status (for example, IL-7R expression) may stand out as superior functional determinants, especially considering that V γ 4⁺V δ 4⁺ $\gamma\delta$ T cells tend to express reduced TCR levels compared to other $\gamma\delta$ T cell subsets³⁷.

Finally, we linked the CCL2-dependent recruitment of $\gamma\delta$ T cells to the TME of M/D-driven carcinomas exposed to P + T with the repolarization of TAMs toward an immunosuppressive CX3CR1⁺ state. TAMs are a major component of the TME in multiple neoplasms, including mouse and human mammary tumors⁶⁷. Unfortunately, currently available CX3CR1-specific antibodies neither block CX3CR1 functions nor deplete CX3CR1⁺ macrophages. With that said, we demonstrated that the immunosuppressive CX3CR1-enriched TME emerging in C57BL/6 mice bearing M/D-driven carcinomas exposed to P + T can be repolarized toward a more immunostimulatory state with antibodies to CSF1R, resulting in superior therapeutic effects than P + T only. Importantly, the pharmacological CSF1R inhibitor pexidartinib is approved for use in individuals with tenosynovial giant cell tumors²⁴. Moreover, pexidartinib as well as antibodies to CSF1R (for example, cabiralizumab) are being tested in individuals with various solid tumors, demonstrating a good safety profile and preliminary signs of efficacy^{68,69}. Our findings suggest that pexidartinib or cabiralizumab may also constitute valid approaches to extend the therapeutic effect of CDK4/CDK6 inhibitors plus endocrine therapy in individuals with advanced/metastatic HR⁺ BC, pending potential safety concerns mentioned above.

Despite these and other unresolved questions, our work delineates an immunosuppressive axis elicited by CDK4/CDK6 inhibitors that offers therapeutically actionable targets to improve the clinical management of HR⁺HER2⁻ BC.

Methods

In vitro experiments with BC cell lines

Primary cell cultures and cell lines. Primary (MGT10 and MGT12) cells were isolated from M/D-driven tumors developed in wild-type C57BL/6J female mice after dissociation in DMEM/F12 (10565042, Fisher Scientific) supplemented with 100 U ml⁻¹ penicillin sodium, 100 μ g ml⁻¹ streptomycin sulfate (10378016, Life Technologies), 100 μ g ml⁻¹ gentamycin (15750078, Life Technologies) and 0.15% type A collagenase (17018029, Thermo Scientific) for 30 min at 37 °C (with occasional mechanical dispersion)²¹. After dissociation, cells were cultured in flasks precoated with 0.1% gelatin (4 h, 37 °C; G1393, Millipore Sigma) in DMEM/F12 medium supplemented with 2% fetal bovine serum (FBS; 900-108, GeminiBio), 50 μ g ml⁻¹ gentamycin, 10 ng ml⁻¹ insulin (I0516, Millipore Sigma) and 5 ng ml⁻¹ epithelial growth factor (SRP3196, Millipore Sigma).

Mouse mammary adenocarcinoma TS/A cells (RRID:CVCL_VQ63) and human mammary adenocarcinoma MCF7 cells (RRID:CVCL_0031) were kindly provided by S. Demaria (Weill Cornell Medicine) and cultured at 37 °C under 5% CO₂ in DMEM supplemented with 4.5 g l⁻¹ glucose (15-017-CV, Thermo Fisher Scientific), 10% FBS, 100 U ml⁻¹ penicillin, 100 μ g ml⁻¹ streptomycin, 0.3 mg ml⁻¹ glutamine (10378016, Life Technologies), 50 μ g ml⁻¹ gentamicin, 5 mM HEPES (15630-080, Life Technologies) and 50 μ M 2-mercaptoethanol (21985023, Life Technologies). All cells were authenticated by STR profiling (a service provided by IDEXX Bioresearch) and routinely checked for *Mycoplasma* spp. contamination by the PCR-based LookOut Mycoplasma PCR Detection kit (MP0035, Millipore Sigma). All cells were used for experiments two to ten passages after thawing.

CCL2 quantification by ELISA. For extracellular CCL2 quantification, 9 × 10⁴ MCF7, 2 × 10³ TS/A cells and 2 × 10⁴ MGT10 and MGT12 cells were seeded in 24-well plates and treated with P (500 nM for TS/A, MGT10 and MGT12 cells and 100 nM for MCF7 cells; HY-A0065, MedChem Express), T (500 nM for TS/A, MGT10 and MGT12 cells and 6 μ M for MCF7 cells; T5648, Millipore Sigma) or their combination. For the quantification of CCL2 release under hypoxic conditions, 6 × 10⁴

MGT10 and MGT12 cells were seeded in six-well plates. The day after, cells were preincubated for 6 h at different oxygen tensions (normoxic conditions: 20% O₂; hypoxic conditions: 1% and 0.01% O₂) and treated at their relative oxygen tension as described above. CCL2 secretion was assessed in the supernatants collected after 72 h of incubation by ELISA with the Mouse CCL2/JE/MCP-1 Quantikine ELISA kit (MJE00B, R&D System) and a Human CCL2/MCP-1 Quantikine ELISA kit (DCP00, R&D System), according to manufacturer's instructions.

In vivo experiments

Mice and oncogenesis. All animal studies were performed following the Guidelines for the Care and Use of Laboratory Animals and under a protocol approved by the Institutional Animal Care and Use Committee (IACUC) of Weill Cornell Medical College (2019–2022). Wild-type C57BL/6J and B6.129P2-*Tcrd*^{tm1Mom}/J (strain 002120)³⁸ female mice (sex determined by relevance for disease type) of 6–7 weeks of age were obtained from The Jackson Laboratories. Mice were housed under standard conditions in line with institutional requirements. Endogenous mammary tumors were established per conventional procedures²¹. In brief, 6- to 7-week-old female C57BL/6J and B6.129P2-*Tcrd*^{tm1Mom}/J mice were implanted subcutaneously with 50 mg of slow-release (90 days) medroxyprogesterone acetate pellets (NP-161, Innovative Research of America). Mice were then administered 1 mg of dimethylbenz[a]anthracene in 200 µl of corn oil (C8267, Millipore Sigma) once a week on weeks 1, 2, 3, 5, 6 and 7 after implantation of the medroxyprogesterone acetate pellet²¹. Mice were routinely monitored for the emergence of neoplastic lesions along the mammary lines, which were allowed to reach a surface area of 15–45 mm² (day 0). Tumor surface was calculated as the area of an ellipse ($A = \text{longest diameter} \times \text{shortest diameter} \times \pi / 4$), as per common procedures²¹. In all animal studies, mice were routinely monitored for tumor growth and killed when the cumulative tumor surface reached 180–200 mm² (the ethical endpoint was used as a surrogate marker for survival and was never exceeded for the purpose of this study) or in the presence of evident signs of toxicity or distress (for example, hunching, anorexia and tumor ulceration), as per common procedures. Drug responses were classified according to RECIST-based criteria into objective response, stable disease and progressive disease based on relative tumor area (A/A_0 ; adapted from Guo et al.²⁹) at day 20 from treatment initiation (objective response: $A/A_0 \leq 0.756$, stable disease: $0.756 < A/A_0 \leq 1.440$, progressive disease: $A/A_0 > 1.440$).

Treatments. On day 0, mice bearing M/D-driven mammary tumors were randomly allocated to the following treatment arms: (1) no treatment, (2) three fractions of 10-Gy focal RT (each on days 0–2), (3) P administered by mouth daily at a concentration of 100 mg per kg (body weight) in 50 mM sodium lactate (L1375, Sigma-Aldrich; pH 4.0, days 0–13), (4) T administered in the drinking water at a concentration of 10 µg ml⁻¹ until the endpoint, (5) P (days 0–13) plus T (from day 0 to the endpoint), (6) focal RT (days 0–2) followed by P (days 3–16), (7) focal RT (days 0–2) followed by T (from day 3 to the endpoint), (8) focal RT (days 0–2) followed by P (days 3–16) plus T (from day 3 to the endpoint) and (9) focal RT (days 0–2) followed by P (days 3–16) plus T (from day 3 to the endpoint) in combination with anti-PD-1 (Supplementary Table 1a) administered intraperitoneally at 0.2 mg per mouse on days 0, 3 and 6. Mice were irradiated with a Small Animal Radiation Research Platform (from Xstrahl). Group sizes are indicated in each figure legend. At least nine mice per group were monitored for tumor growth and OS, as described above. For scRNA-seq experiments, three mice per group were treated with the above treatments from days 0 to 5, and tumors were collected at day 6 and processed as described below.

Experimental procedures to control for potential batch-dependent effects. The following procedures were implemented to control for batch-dependent effects that may potentially emerge from the

intrinsically longitudinal nature of accrual to experimental groups imposed by the M/D-driven model²¹. First, while accrued longitudinally over various batches of M/D-driven carcinogenesis established 3–4 months apart from each other, experimental groups involved in direct statistical comparisons were accrued in parallel in a step-wise manner based on two to four mice per batch until completion of the group. Second, each batch of M/D-driven carcinogenesis was conceived to include four to six untreated mice that were compared for tumor growth and OS to historical data from mice with untreated M/D-driven mammary carcinomas compiled since the establishment of the model, and the entire batch was discarded in case either of these parameters significantly differed from existing data (see below for statistical tests used). Third, the validity of all statistical assessments on tumor growth and OS involving untreated M/D-driven tumors and M/D-driven tumors receiving P + T presented in the current paper was confirmed with four extra groups of untreated mice ($n = 10$ per batch) and one extra group of P + T-treated mice ($n = 12$), as well as with two large sets including these extra groups plus the groups of untreated mice ($n = 14$) and P + T-treated mice ($n = 15$) presented in the paper (untreated mice: $n = 54$; P + T-treated mice: $n = 27$).

In vivo depletion and neutralization. All antibodies blocking or neutralizing specific immune cell populations or cytokines are reported in Supplementary Table 1a. Mice were treated by intraperitoneal injection with 0.1 mg of anti-CCL2 every other day and with 0.1 mg of anti-IL-17A or anti-TCRγ/TCRδ twice a week. Anti-CSF1R was administered intraperitoneally in the amount of 1.2 mg per mouse (as a first dose), followed by 0.6 mg per mouse twice a week (days 0–13) and then once a week until the endpoint. For co-depletion of CD4⁺ and CD8⁺ T cells, mice were treated by intraperitoneal injection with 0.4 mg of anti-CD4 and 0.2 mg of anti-CD8 once every 2 weeks.

Phenotyping of immune cells infiltrating M/D-driven tumors by flow cytometry. M/D-driven tumors treated as described above were allowed to reach the endpoint. Tumor tissues were excised and homogenized into single cells using a mouse tumor dissociation kit (130-096-730, Miltenyi) and with a gentleMACS Octo Dissociator (Miltenyi). Erythrocytes were removed by incubating digested tumors in 1× red blood cell lysis buffer (420301, BioLegend; vol/vol distilled water) for 5 min. Single-cell suspensions were then passed through a 70-µm cell strainer (1164H36, Thomas Scientific) and subjected to magnetic separation of CD45⁺ cells using CD45 (TIL) MicroBeads (130-110-618, Miltenyi) and a MACS Separator (Miltenyi), as per the manufacturer's instructions. Two million CD45⁺ cells were stained as follows: (1) with antibodies to CD45, CD3ε, CD8, CD4, TCRγ/TCRδ and CD335/NKp46 for lymphocyte profiling; (2) with antibodies to CD3ε, CD8, CD4, TCRγ/TCRδ, IL-17A and IFNγ, after ex vivo stimulation for 3 h with PMA plus ionomycin (1:500; 00497093, Fisher Scientific) and brefeldin A (1:1,000; 420601, BioLegend), for the characterization of cytokine secretion by T cells and (3) with antibodies to CD80, MHC class II (I-A/I-E), CX3CR1, F4/80 and CD45, after ex vivo polarization for 24 h with IFNγ (50 ng ml⁻¹; 315-05, Thermo Fisher) plus lipopolysaccharide (100 ng ml⁻¹; 00-4976-93, Thermo Fisher), for macrophage profiling. A Zombie Aqua Fixable Viability kit (423102, BioLegend) was used for all three panels to identify live cells. Cells were incubated with the aforementioned antibodies for 30 min at 4 °C in the dark and permeabilized with eBioscience permeabilization buffer (part of 88-8824-00, Thermo Fisher) for intracellular staining (IL-17A and IFNγ). Catalog reference numbers and dilutions for all antibodies are reported in Supplementary Table 1a. Flow cytometry was performed using a MACSQuant Analyzer 10 operated by MACSQuantify v.2.11 (Miltenyi) or an Attune NxT Flow Cytometer operated by Attune NxT Cytometer Software v.5.3 (Thermo Fisher), and data were analyzed with FlowJo v. 9.5.3 (TreeStar). Gating strategies are shown in Supplementary Figs. 1 and 2.

Quantification of circulating cytokines in M/D-driven tumor-bearing mice. Mice bearing M/D-driven tumors were left untreated or treated with RT, P + T or their combination, as described in ‘Treatments’, for 6 days (days 0–5). Blood was collected at days 0 and 6. After collection, whole blood was allowed to clot at room temperature for at least 30 min. Plasma was isolated after removal of the clot with centrifugation at 2,000g for 10 min, and circulating cytokines were quantified by flow cytometry with the LEGENDplex Mouse Inflammation Panel (740446, BioLegend). All samples were analyzed in technical duplicate. Data were acquired with a Fortessa analyzer (BD Biosciences) and further analyzed using the LEGENDplex Data Analysis Software Suite (<https://legendplex.qognit.com>).

Quantification of CCL2 in M/D-driven tumors. Mice bearing M/D-driven tumors were left untreated or treated with P, as described above, for 6 days (days 0–5). At day 6, M/D-driven mammary tumors were collected and lysed with 1% SDS lysis buffer (0.1 M Tris (pH 8.0), 1% SDS and 10% glycerol in water) on ice for 10 min, followed by boiling for 5 min. Immunoblotting was performed according to conventional procedures, with primary antibodies specific to β -actin and MCP-1, respectively (Supplementary Table 1a). After washing and incubation with horseradish peroxidase-conjugated anti-mouse secondary antibody from Cytiva (Supplementary Table 1a), visualization was performed with SuperSignal West Femto Maximum Sensitivity Substrate (34094, Thermo Fisher) on an Azure 600 Ultimate Western Imaging System operated by Azure software v.1.9.0.0406 (Azure Biosystems).

Detection of RT-induced hypoxia in M/D-driven tumors by immunofluorescence. Mice bearing M/D-driven tumors were left untreated or treated with focal RT or P, as described above, for 6 days (days 0–5). At day 6, mice bearing M/D-driven mammary tumors were intraperitoneally injected with 65 mg per kg (body weight) pimonidazole HCl (HP8, Hypoxyprobe) in PBS 1.5 h before being killed. Tumor tissues were directly embedded in Tissue-Tek O.C.T. compound (4583, Sakura Finetek). Ten-micron-thick frozen sections were fixed with 4% paraformaldehyde (sc-281692, Santa Cruz Biotechnology) for 10 min at room temperature, incubated with 0.5% bovine serum albumin (BSA; 700-100P, GeminiBio) plus 0.3% Triton X-100 (X100, Sigma-Aldrich) in PBS for 10 min and incubated overnight at 4 °C with anti-pimonidazole (Supplementary Table 1a) in PBS plus 0.5% BSA and 0.3% Triton X-100. After washing with 0.5% BSA/PBS, incubating with 2 $\mu\text{g ml}^{-1}$ Hoechst 33258 (H3569, Thermo Fisher) and mounting with ProLong Glass Antifade Mountant (P36984, Thermo Fisher), images were acquired with an LSM 880 confocal laser-scanning microscope operated by ZEN Black v. 2.3 SPI (Zeiss). Quantification was performed with ImageJ v1.53t (<http://imagej.nih.gov/ij>) to measure and calculate the percentage of pimonidazole⁺ area.

Ex vivo experiments

Splenocyte sensitivity assays. To determine the sensitivity of $\gamma\delta$ T and non- $\gamma\delta$ T cells to hypoxia, 10^6 healthy splenocytes were plated in 24-well plates precoated with 5 $\mu\text{g ml}^{-1}$ anti-CD3 (14-0031-85, eBiosciences) and 50 U ml^{-1} soluble recombinant mouse IL-2 (212-12-5ug, PeproTech) and incubated under different oxygen tensions (20%, 5% and 1% oxygen). After 72 h of incubation, cells were collected and labeled with a Zombie Aqua Fixable Viability kit and antibodies specific to CD3 ϵ , CD8, CD4 and TCR γ /TCR δ (Supplementary Table 1a), and the percentage of live $\gamma\delta$ T and non- $\gamma\delta$ T cells was analyzed by flow cytometry.

scRNA-seq analysis of M/D-driven tumors

Generation of single-cell suspensions. Single-cell suspensions from M/D-driven tumors were generated as described above and immediately subjected to magnetic separation of CD45⁺ cells. CD45⁺ fractions were processed for the enrichment of malignant (CD45⁺ EpCAM⁺) and nonmalignant (CD45⁺ EpCAM[−]) cells by magnetic separation using

CD326 (EpCAM) MicroBeads (130-105-958, Miltenyi). A small aliquot of enriched CD45⁺, CD45⁺ EpCAM⁺ and CD45⁺ EpCAM[−] cells was stained with anti-CD45-FITC and anti-CD326-AF647 (Supplementary Table 1a) to determine the purity of each fraction and estimate the number of cells by flow cytometry. Each fraction was immediately resuspended in 1 ml of Cryostore CS10 (07930, Stemcell Technologies), and cells were cryopreserved using a slow rate-controlled cooling container (approximately $-1\text{ }^{\circ}\text{C min}^{-1}$) and stored in liquid nitrogen ($-135\text{ }^{\circ}\text{C}$) after 24 h.

scRNA-seq library generation, sequencing and data preprocessing.

Cryopreserved cells were gently thawed using a 37 °C water bath and washed with prewarmed growth medium (10% FBS in DMEM). Cells were washed twice with FACS buffer (10 μM pH 8 EDTA + 5% heat-inactivated FBS + 0.001% (mass/vol) NaN_3 in PBS) and stained with Biolegend HTO antibodies, following standard manufacturer procedures, and counted with a hemocytometer. Cells from approximately eight samples were then pooled in equal proportions, and the samples were processed according to the 10x Genomics protocol for the Single Cell 3' v2 assay with a targeted 10,000 cell recovery. Subsequent gene expression and HTO libraries were prepared according to the 10x Genomics Single Cell 3' v2 protocol. Demultiplexed fastq files for the gene expression libraries were processed with Cell Ranger (v4.0). Demultiplexed fastq files for the HTO libraries were processed with CITE-Seq-Count (v1.4.3) using the reported number of cells from the Cell Ranger output, the filtered cell barcode list from the Cell Ranger output and the appropriate HTO tag sequences to generate HTO count (unique molecular identifier (UMI)) tables. These HTO count tables were then processed with GMM-Demux, again using the reported number of cells from the Cell Ranger output and a threshold of 0.3, to identify singlets and multiplets. All further data processing steps were performed using scripts in the Python 3.8 environment. Cells classified as multiplets by GMM-Demux were further screened by assessing the ratio of the counts for the highest to second-highest expressed HTO tags. Cells where this ratio was greater than a factor of 5 (that is, the highest expressed tag had more than five times the number of counts than the second-highest expressed tag) were reclassified as singlets. All remaining multiplets and nonclassified cells were removed from further analysis. Singlets were further filtered to express at least 250 unique genes, contain less than 30,000 total UMIs and have less than or equal to 10% of total UMIs associated with mitochondrial genes. Dimensional reduction and clustering were performed on total count-normalized, log-transformed counts using Scanpy's (v1.5.1) implementation of the self-assembling manifold method ($k = 30$, distance = euclidean, standardization = Normalizer) and Leiden clustering (resolution = 0.8), respectively. DGEA was performed using `sc.tl.rank_genes_groups` (method = wilcoxon), and gene signature scores were generated using `sc.tl.score_genes` on total count-normalized, log-transformed counts.

TCR-sequencing analysis of M/D-driven tumors. Total RNA was isolated using an RNeasy mini kit (74104, Qiagen) from M/D-driven tumors left untreated or treated with P + T as described in ‘Treatments’ and collected at the endpoint. Libraries were prepared using a Qiaseq Targeted RNA Panel Mouse TCR kit (334671, Qiagen). Demultiplexed fastq files were deduplicated and aligned using MiXCR with the Qiaseq preset and the following tag pattern flag: `--tag-pattern ^{(R1:*)} \ ^{(UMI:N{12:18})} N{12}(R2:*)`. Resulting clonotype tables were filtered to only productive TCR alignments and used for downstream analysis in Python.

Human data

METABRIC survival analysis. Gene expression data for the METABRIC cohort³⁹ were downloaded from cBioPortal (https://www.cbioportal.org/study/summary?id=brca_metabric) on 18 November 2020, and extended clinical data were obtained from Rueda et al.⁷⁰. GO gene sets for the GO:0032740 signature were retrieved from the Molecular Signatures Database⁷¹ using the msigdb 7.4.1 R package. Cell-type

deconvolution was performed with xCell to calculate the abundance of $\gamma\delta$ T cells. Single-sample scores were calculated with GSVA1.42.0 for GO gene sets and with xCell for cell-type signatures using the immune-deconv 2.0.4 R package. Participants were separated into ER⁺ and ER⁻ subgroups based on clinical annotation. Participants in the complete cohort and in the ER⁻ defined subgroups were further stratified by median values of single-sample scores. The survival 3.2-13 and survival 0.4.9 R packages were used for survival analysis. All analyses were performed in R 4.1.0.

Participant cohorts. The data from this study were obtained from five different cohorts.

Cohort 1. Biopsies from 84 participants with luminal A (HR⁺HER2⁻; grade 1: $n = 16$, grade 2: $n = 55$ and grade 3: $n = 13$) BC were obtained at the Institute of Pathology, University Hospital Halle (Saale), Germany. Written informed consent to participate in this study was obtained from all participants in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013)⁷². The study protocol was approved by the Institutional Review Board of University Hospital Halle (2017-81). Luminal B molecular phenotype was determined by immunohistochemistry criteria according to St. Gallen International Guidelines⁷³. See also Supplementary Table 2.

Cohort 2. From 2017 to 2021, 12 women with high-risk luminal A HR⁺HER2⁻ early-stage BC were enrolled in a clinical trial of preoperative pembrolizumab-mediated immune therapy with radiation and serial tissue collection (NCT03366844) at Cedars-Sinai Medical Center (Los Angeles, California)⁴⁶. Written informed consent to participate in this study was obtained from all participants in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013)⁷². The study protocol was approved by the Institutional Review Board of Cedars-Sinai Medical Center (Pro00049440). High-risk luminal A BC was defined by the presence of at least two of the following: histologic grade of 2–3, Ki67 of >20% and/or ER expression of <75% by immunohistochemistry⁷⁴. For each participant, biopsies were collected before treatment initiation, at 4 weeks after treatment initiation and at surgical resection. See also Supplementary Table 3.

Cohort 3. Twenty-nine participants with luminal A and B advanced BC were recruited at Virgen Macarena University Hospital (Seville, Spain). Written informed consent to participate in this study was obtained from all participants in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013)⁷². The study protocol was approved by the Institutional Review Board of Virgen Macarena University Hospital (LCM-INM-2015–01/Law 14/2007, of July 3). Participants received CDK4/CDK6 inhibitors as indicated in local protocols based on International Guidelines, including National Comprehensive Cancer Network (NCCN)⁷⁵ and advanced BC Guidelines⁷⁶. Luminal B molecular phenotype was obtained by immunohistochemistry criteria according to St. Gallen International Guidelines⁷³, and clinical response was assigned following the revised RECIST v1.1 to consider participants who achieve clinical benefits (complete response, partial response or stabilization of disease) and participants who experience disease progression to CDK4/CDK6 inhibitors. Peripheral blood samples were collected in EDTA-K3 tubes at baseline, after three cycles of treatment (re-evaluation) and after six cycles of treatment. See also Supplementary Table 4.

Cohort 4. Twenty-eight participants with advanced luminal A and B BC were recruited and treated at University Hospital Morales Meseguer (Murcia, Spain). Written informed consent to participate in this study was obtained from all participants in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013)⁷². The study protocol was approved by the Institutional Review Board

of University Hospital Morales Meseguer (EST24/15). Participants received CDK4/CDK6 inhibitors and endocrine therapy according to local protocols based on NCCN and advanced breast cancer (ABC) management Guidelines⁷⁶ and were included at the beginning of the treatment in the study HMM-11/18 from June 2018 to May 2021. Response evaluation was performed according to RECIST 1.1 criteria, with 66.7% of participants experiencing disease progression to CDK4/CDK6 inhibitors and endocrine therapy. Two EDTA-K2 10-ml tubes of whole blood were collected from each participant at the beginning of the treatment, after 3 months (re-evaluation) and at disease progression. Plasma and peripheral blood mononuclear cells were obtained and frozen at –80 °C until analysis. See also Supplementary Table 4.

Cohort 5. Sixty-six participants diagnosed with advanced luminal BC received CDK4/CDK6 inhibitors at the Clínica Universidad de Navarra (Pamplona, Spain). Written informed consent to participate in this study was obtained from all participants in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013)⁷². The study protocol was approved by the Institutional Review Board of Clínica Universidad de Navarra (2010.111 mod4). Nineteen participants with matching tumor biopsies obtained before and after CDK4/CDK6 inhibitors were found, but only eight were selected based on tissue quality. See also Supplementary Table 4.

Immunohistochemistry. Immunohistochemistry staining was performed on formaldehyde-fixed paraffin-embedded (FFPE) BC biopsies obtained from 86 participants with HR⁺HER2⁻ BC of different grades (grade 1 ($n = 16$), grade 2 ($n = 57$) and grade 3 ($n = 13$); Cohort 1) by using a Bond III automated immunostainer (Leica Biosystems) and a Bond Polymer Refine Detection kit (DS9800-CN, Leica Biosystems). Primary anti-CD68, anti-CD8 and anti-TCR δ were used as reported in Supplementary Table 1b. Immunostaining was assessed blinded to additional pathological and clinical data using a Zeiss Axioscope 5 microscope (Carl Zeiss Microscopy). Semiquantitative evaluation was performed using the Histo-scoring system.

Multispectral immunofluorescence. Data were obtained from two independent cohorts.

(1) Multispectral immunofluorescence for the simultaneous detection of TCR $\gamma\delta$, CD68 and PD-L1 was performed on FFPE BC biopsies obtained from 26 participants with HR⁺ BC of different grades (grade 1 ($n = 8$), grade 2 ($n = 18$); Cohort 1) using an Opal 7-Color Manual IHC kit (NEL811001KT, Akoya Bioscience) and the following antibody–dye combinations: anti-TCR δ (pH 9) with Opal 540, anti-CD68 (pH 9) with Opal 570 and anti-PD-L1 (pH 6) with Opal 690, followed by counterstaining with DAPI (FP1490, Akoya Bioscience). Blocking was performed by implementing reagents from the Opal kit. Slides mounted with antifade medium (VEC-H-1000, Vectashield) were evaluated on a Vectra Polaris microscope (Akoya Bioscience). A mean of 9 different spots (range: 2–20) per slide were imaged and evaluated using InForm software and the phenoptr package.

(2) A multiplex immunofluorescence protocol was developed for simultaneous detection of TCR $\gamma\delta$, CD8, CD4, FOXP3, PD-L1, pan-CK and DAPI in a single FFPE tissue section, following a previously described validation pipeline⁷⁷, and was applied to matching tumor biopsies obtained before and after treatment from eight individuals with HR⁺ BC (Cohort 5). Full details of antibodies are provided in Supplementary Table 1b. Spectral unmixing was performed using InForm software (v2.4.8, Akoya Biosciences), as previously described⁷⁸. Image analysis was performed using QuPath (v0.2.3, <https://qupath.github.io>) and Fiji/ImageJ (v1.52c, <https://imagej.net/ij>). Briefly, cells were segmented based on nuclear detection using the StarDist 2D algorithm. A random trees algorithm classifier was trained for each cell marker. Cells were then subclassified as $\gamma\delta$ T cells, CD8⁺ T cells, CD4⁺ T cells and T_{reg} cells (CD4⁺FOXP3⁺). Subpopulations within T cells were then classified

by the presence and absence of PD-1. Pan-CK was used to visualize tumor cells. Cells negative for these markers were defined as 'other cell types'. Measurements were calculated as cell densities (cells mm⁻²). Ulcerative colitis tissue sections served as positive tissue controls (Extended Data Fig. 7c).

Phenotyping of circulating T cells by flow cytometry. Data were obtained from two independent cohorts.

(1) For the profiling of circulating CD4⁺CD8⁺ T cells in participants with HR⁺ BC from Cohort 3, whole-blood samples collected at baseline ($n = 29$) were stained with antibodies specific to CD3, CD4 and CD8 (Supplementary Table 1b) for 15 min at room temperature in the dark, after removal of erythrocytes by BD FACS Lysing Solution (1×; 349202, BD Biosciences). Samples were analyzed with a BD FACSCanto II flow cytometry system (Becton Dickinson), and data were analyzed using FlowJo v. 9.5.3 (TreeStar). The gating strategy is shown in Supplementary Fig. 3.

(2) For the profiling of circulating CD4⁺CD8⁺ T, CD8⁺PD-1⁺ T, T_{reg} and IL-7R⁺ γδ T cells in participants with HR⁺ BC from Cohort 4, 3×10^5 peripheral blood mononuclear cells obtained from blood samples collected at baseline ($n = 23$), after 3 months from the start of treatment ($n = 23$) and at disease progression ($n = 14$) were treated with beriglobine P (FcRs, anti-CD16, CD32, CD64; 18091A, CSL Behring) in 10% FBS/PBS buffer and stained with PromoFluor 840 (PK-PF840-3-01, Promocell) and antibodies specific to CD45, CD3, CD56, TCRγ/TCRδ, CD8-APC, CD4, CD25, PD-1, CD11b, CD19, CD127/IL-7Rα and FOXP3 (Supplementary Table 1b). Before intracellular FOXP3 staining, cells were fixed and permeabilized using a True-Nuclear Transcription Factor Buffer Set (424401, Biolegend). Samples were acquired on a CytoFLEX flow cytometer (BD Biosciences), and analysis was performed using Cytexpert software (v2.4). The gating strategy is shown in Supplementary Fig. 4. Absolute numbers of circulating cells were calculated by multiplying the percentages obtained from flow cytometry by total lymphocyte count obtained from the hematologic count (Sysmex CS-1000).

Quantification of human circulating CCL2. For the quantification of serum CCL2 levels in participants with HR⁺ BC (Cohort 4), blood samples were collected as described above. CCL2 levels were quantified by flow cytometry using a BD Cytometric Bead Array (558264, BD Biosciences) and FlowJo v. 9.5.3 (TreeStar) using a downloadable plugin (CBA kit analysis), which generated clustering of each sample according to FSC versus SSC morphological parameters and calculated the relative analyte concentration (based on mean fluorescence intensity (MFI)) based on the uploaded Cytometric Bead Array standard. All samples were analyzed in technical duplicate.

Human scRNA-seq library construction, sequencing and data preprocessing. Library construction, sequencing and data preprocessing on human samples were performed as previously reported⁷⁹. Specifically, as reported in that paper: 'Core biopsies were obtained under local anesthesia using a 12 G core needle apparatus (Bard Magnum). Biopsies were placed immediately into a 50 mL conical tube containing ice-cold DMEM and transferred on ice from the clinic to the laboratory (generally within 30 min of the procedure). A portion of the biopsy was immediately placed into formalin for subsequent paraffin embedding. The remainder of the biopsy was minced with a sterile scalpel into 2 mm pieces and placed in a gentleMACS™ C-tube (Miltenyi Biotech) containing 4.7 mL of ice-cold DMEM. Enzymes from the Human Tumor Dissociation Kit (Miltenyi Biotech) were added to the sample according to the manufacturer's protocol for dissociation of tough tumors. After the enzymes were added, the C-tube was placed onto a gentleMACS Octo Dissociator (Miltenyi Biotech), the heating element was placed over the tube, and the 37C_h_TDK_3 program was initiated. Prior to completion of the dissociation protocol, two 70 μm cell strainers

were each placed onto a 50 mL conical tube on ice and pre-wetted with 3 mL of heat-inactivated FBS. Upon completion of the dissociation protocol, the sample was poured through the first 70 μm cell strainer and the strainer was washed with 5 mL of heat-inactivated FBS. The filtrate was then poured through the second 70 μm cell strainer and the strainer was washed with 5 mL of heat-inactivated FBS. The sample was centrifuged at 300 g for 7 min at 4 °C. The supernatant was removed, and the sample was resuspended in 2 mL of DMEM and transferred to a microcentrifuge tube on ice. An aliquot of the sample was taken for cell counting via a hemocytometer and the remaining sample was centrifuged at 300 g for 10 min at 4 °C. The supernatant was removed, and the sample was resuspended in 80 μL FACS buffer (10 μM pH 8 EDTA + 5% H.I. FBS + 0.001% m/v NaN₃ in PBS, sterile filtered) per 10⁷ cells. Miltenyi anti-human CD45 microbeads were added to the sample (20 μL per 10⁷ cells), and the sample was incubated at 4 °C covered with foil for 15 min. After incubation, 1 mL FACS buffer per 10⁷ cells was added to the sample and pipet mixed. The sample was centrifuged at 300 g for 10 min at 4 °C. The supernatant was removed, and the sample was resuspended with 500 μL FACS buffer and transferred to a 15 mL conical. The sample was placed in a pre-cooled AutoMACS tube holder and the Miltenyi AutoMACS Possel_S program was initiated. At this time, 10X Genomics reagents required for running the Single Cell 5' v1 Assay were removed from their storage conditions to equilibrate at room temperature for 30 min. Upon completion of the AutoMACS program, the CD45-positive fraction was centrifuged at 300 g for 7 min at 4 °C. The sample was resuspended in 50 μL of DMEM and a cell count was performed via a hemocytometer. If possible, the sample was diluted to 10³ cells/μL and the sample was processed according to the 10X Genomics protocol for the Single Cell 5' v1 assay with a targeted 10,000 cell recovery (or maximum recovery possible with the number of cells in the sample). Subsequent gene expression libraries [...] were prepared according to the 10X Genomics Single Cell 5' v1 protocol. Gene expression libraries were sequenced using paired end 150 bp reads to an average 60-70% sequencing saturation (as calculated by 10X Genomics Cellranger v4.0.0) on an Illumina Novaseq. [...] Demultiplexed fastq files for gene expression libraries were processed using Cellranger (v4.0.0, GRCh38-3.0.0 transcriptome reference, read 1 and read 2 set to 100 bp). Doublet cells were identified using Scrublet from the filtered feature barcode matrices produced by Cellranger. Singlets were further filtered to express at least 250 unique genes, contain less than 30×10^5 total UMIs and have ≤10% total UMIs associated with mitochondrial genes.

Statistical analysis and reproducibility

Unless otherwise specified, data management, analysis and graphing were performed with Prism v. 8.4 (GraphPad) or Excel 365 ProPlus (Microsoft). Figures were prepared with Illustrator 2023 (Adobe). Unless otherwise indicated, in vitro and ex vivo results were obtained from at least three independent biological samples collected over at least two independent experiments.

For statistical analysis, the normality of the distributions was assessed using a Shapiro–Wilk test.

Statistical significance of CCL2 release by BC cell lines, splenocyte sensitivity to hypoxia and pimonidazole⁺ area in M/D-driven tumors were assessed by linear mixed effects regression while accounting for potential within-experiment correlations. Heteroscedasticity in within-group errors was modeled by allowing different variances for different groups through group-specific weights. Simultaneous tests for generalized linear hypotheses were used to evaluate contrasts of interest. *P* values were not adjusted for multiple comparisons.

To assess differences in tumor growth of M/D-driven mammary tumors, linear mixed effects models with treatment group, time and treatment group by time interaction as the fixed effects and random intercept and slope by mouse were applied to model primary and cumulative tumor size. Tumor sizes were square root transformed to

ensure that the underlying model assumptions were satisfied. Heterogeneity in variances were modeled by applying time-dependent weights. Tumor growth rates were compared for contrasts of interest using simultaneous tests for general linear hypotheses. *P* values were not adjusted for multiple comparisons.

Statistical significance of survival of M/D-driven-tumor-bearing mice was assessed by a two-sided Gehan–Breslow–Wilcoxon test or log-rank test. Statistical significance of OS, disease-specific survival and relapse-free survival of participants in the METABRIC database and other cohorts was assessed by log-rank test.

Statistical significance of TCR δ , CD68 and CD8 positivity as determined by immunohistochemistry of human breast tumors was assessed by Kruskal–Wallis test, followed by a Dunn’s test. Statistical significance of mouse flow cytometry data (that is, circulating levels of IL-17A in M/D-driven-tumor-bearing mice, the abundances of T cell populations and macrophages infiltrating M/D-driven tumors and MHC class II and CD80 expression on CX3CR1⁺ macrophages) was assessed by one-way ANOVA plus Fisher’s LSD or Kruskal–Wallis test with an uncorrected Dunn’s test. Statistical significance between paired samples was assessed by two-tailed paired *t*-test or Wilcoxon matched-pairs signed rank test. Differences in CCL2 levels in M/D-driven tumors were assessed by two-tailed unpaired *t*-test.

Spearman’s rank correlation was estimated to determine the linear association between circulating levels of cytokines in mice bearing M/D-driven tumors and between circulating immune cells in longitudinal blood samples collected from participants with HR⁺ BC.

Statistical significance of the abundance of circulating levels of CCL2 or IL-17 γ δ T cells in participants with HR⁺ BC was determined by two-sided Wilcoxon matched-pairs (for differences between longitudinal samples from the same participant) or unpaired (for differences between participants with relapsed disease and those with nonrelapsed disease) signed rank test.

Mann–Whitney–Wilcoxon tests were used to determine statistical differences in ‘cellular response to IL-17’ GO signature scores between CX3CR1^{hi} and CX3CR1^{lo} macrophages, and a Mann–Whitney–Wilcoxon test (two-sided and FDR-adjusted (Benjamini–Hochberg)) was used to determine statistical differences in ‘ γ δ T cells’ and ‘Positive regulation of IL-17 production’ signature scores in ER⁺ breast tumors of different grades from the METABRIC dataset. Limma 3.50.0 was used to test for differentially expressed genes between IL-17^{hi} and IL-17^{lo} participants with ER⁺ BC from the METABRIC database.

For human and mouse single-cell sequencing data, single cells were batch corrected and projected into a low-dimensional representation by single-cell variational inference (scVI; scvi-tools v0.6.0). The scVI model was first trained using 2,000 highly variable genes detected in at least 10% of cells across the whole dataset (scanpy). Using 64 latent dimensions and two hidden layers with 256 nodes a piece, the model was trained until the validation loss failed to decrease for five consecutive epochs. The latent space representation for each cell was extracted from the trained model and processed with UMAP (cuML; v22.02) to produce a two-dimensional representation of the full latent space and to identify unsupervised clusters of cells within the latent space with the Leiden algorithm (cuML). γ δ T cells were annotated in the human data using the Celltypist algorithm. GSEA on human γ δ T cells and CX3CR1⁺ TAMs was performed with the GSEAPy Python package. Statistical differences between CD3⁺ T cells and macrophage clusters identified by mouse scRNA-seq were assessed by a full pairwise χ^2 statistical test.

For in vivo studies, group sizes of 9–15 mice were chosen based on previous experience, considering the heterogeneity of the model^{16,21,25,80} and following the IACUC guidelines from Weill Cornell Medical College. Mice were randomly assigned to different treatment groups, except for mice that received focal radiotherapy, alone or preceding systemic therapy. For these animals, only those with a single treatable primary lesion at day 0 were selected as the presence of multiple

primary nonirradiated tumors at day 0 would have confounded the interpretation of the results.

Investigators were generally not blinded to allocation during in vivo, ex vivo and in vitro experiments and the preliminary analysis thereof, but blinding was implemented on statistical assessments of bulk as well as scRNA-seq studies, as well as all studies using human samples.

No exclusion criteria were pre-established in our analysis. No data points meeting technical quality standards were excluded from the analyses.

Only female mice and female participants were involved in these studies because BC predominantly affects women.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Mouse scRNA-seq and TCR-seq data have been deposited in Gene Expression Omnibus (GEO) under accession number [GSE296798](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE296798). Human scRNA-seq data have been deposited as raw count files in GEO under accession number [GSE296805](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE296805). Human scRNA-seq raw fastq files have not been made publicly available due to privacy concerns but can be made available from the corresponding authors upon request. All requests will be fulfilled within 2 weeks, with data being made available permanently to the requester. Publicly available transcriptional data from the METABRIC dataset^{39,70} were downloaded from cBioPortal (https://www.cbioportal.org/study/summary?id=brca_metabric) on 18 November 2020. The remaining data supporting the findings of this study are available within the article, Supplementary Information or Source Data files and/or from the corresponding authors upon request. Source data are provided with this paper.

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Author contributions

L.G., S.C.F., S.R.V.K. and H.L.M. conceived the project and provided senior supervision to the study. G.P. and C.G. performed preclinical experiments as well as analysis on circulating CCL2 in participants, with support from A.B., N.B., T.Y., A. Sato, M.B.-V. and G.C., under supervision from L.G. K.H.G., H.-H.C. and A. Shah performed and analyzed the RNA-seq experiments under supervision from S.R.V.K. C.J.-C. and I.S.-M. performed and analyzed multicolor flow cytometry experiments on human blood samples under supervision from V.S.-M. and L.d.I.C.-M. or E.G.-M., F.A.d.I.P. and M.E.R.-R., respectively. A.K. analyzed public human datasets under supervision from F.F. and Z.T. C.M. and C.W. performed immunofluorescence microscopy and immunohistochemistry experiments on human tissue samples, respectively, under supervision from B.S. C.E.d.A. performed multispectral immunofluorescence on human tissue samples under supervision from M.E.R.-R. B.N.-R. retrieved deidentified clinical data on human samples under supervision from M.E.R.-R. E.N.M. retrieved deidentified clinical data on human samples under supervision from F.A.d.I.P. and E.G.-M. A.M.S. performed hypoxia assays under supervision from J.M.S. B.R. and X.K.Z. provided statistical support to the project. R.K.B. and S.L.S. ran clinical study [NCT03366844](#) and provided samples for scRNA-seq data. G.P., C.G. and L.G. wrote the paper with critical input from all authors. G.P. and C.G. prepared figures under supervision from L.G. All authors approved the submitted version of the article.

Competing interests

F.F. has been holding research contracts with iOnctura. M.E.R.-R. reports receiving research funding from Roche and Bioncotech. She also has received speaker's bureau honoraria from Bristol Myers Squibb (BMS) and Roche. H.L.M. is/has been holding research contracts with BMS, MedImmune, LLC/AstraZeneca, BTG and Merck and has received consulting/advisory honoraria from Amgen, BMS, Celgene, Eli Lilly, Genentech/Roche, Immunomedics, Merck, OBI Pharma, Pfizer, Puma, Spectrum Pharmaceuticals, Syndax Pharmaceuticals, Peregrine, Calithera, Daiichi-Sankyo, Seattle Genetics, AstraZeneca, Gilead, Crown Bioscience and TapImmune. S.C.F. is/has been holding research contracts with Merck, Varian, BMS, Celldex, Regeneron, Eisai and Eli Lilly and has received consulting/advisory honoraria from Bayer, BMS, Varian, Elekta, Regeneron,

Eisai, AstraZeneca, MedImmune, Merck US, EMD Serono, Accuray, Boehringer Ingelheim, Roche, Genentech, AstraZeneca, View Ray and Nanobiotix. S.R.V.K. is a founder and consultant at Transomic Technologies and Faeth Therapeutics. L.G. is/has been holding research contracts with Lytix Biopharma, Promontory and Onxeo, has received consulting/advisory honoraria from Boehringer Ingelheim, AstraZeneca, OmniSEQ, Onxeo, The Longevity Labs, Inzen, Imvax, Sotio, Promontory, Noxopharm, EduCom and the Luke Heller TECPR2 Foundation and holds Promontory stock options. All other authors have no conflicts to declare.

Additional information

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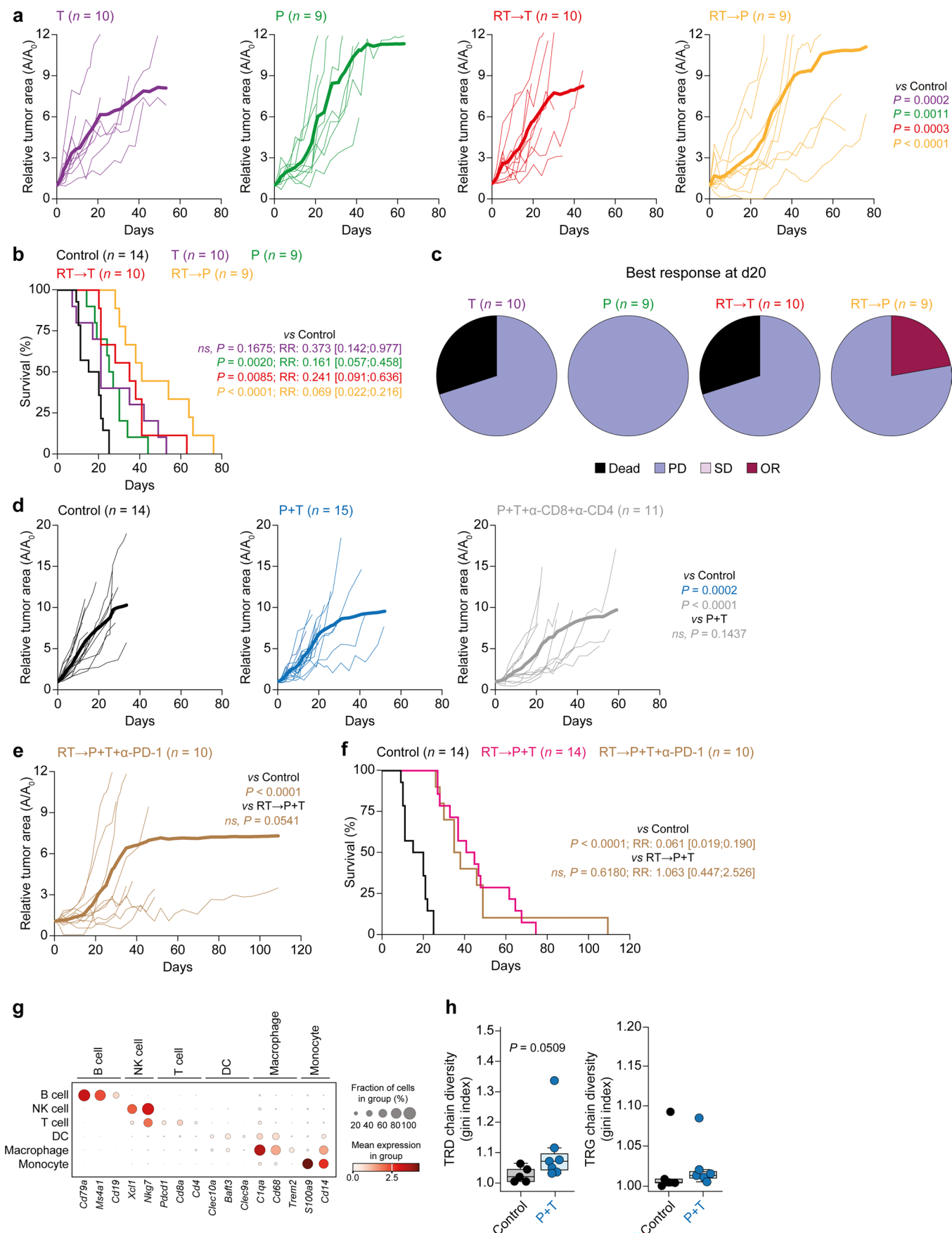
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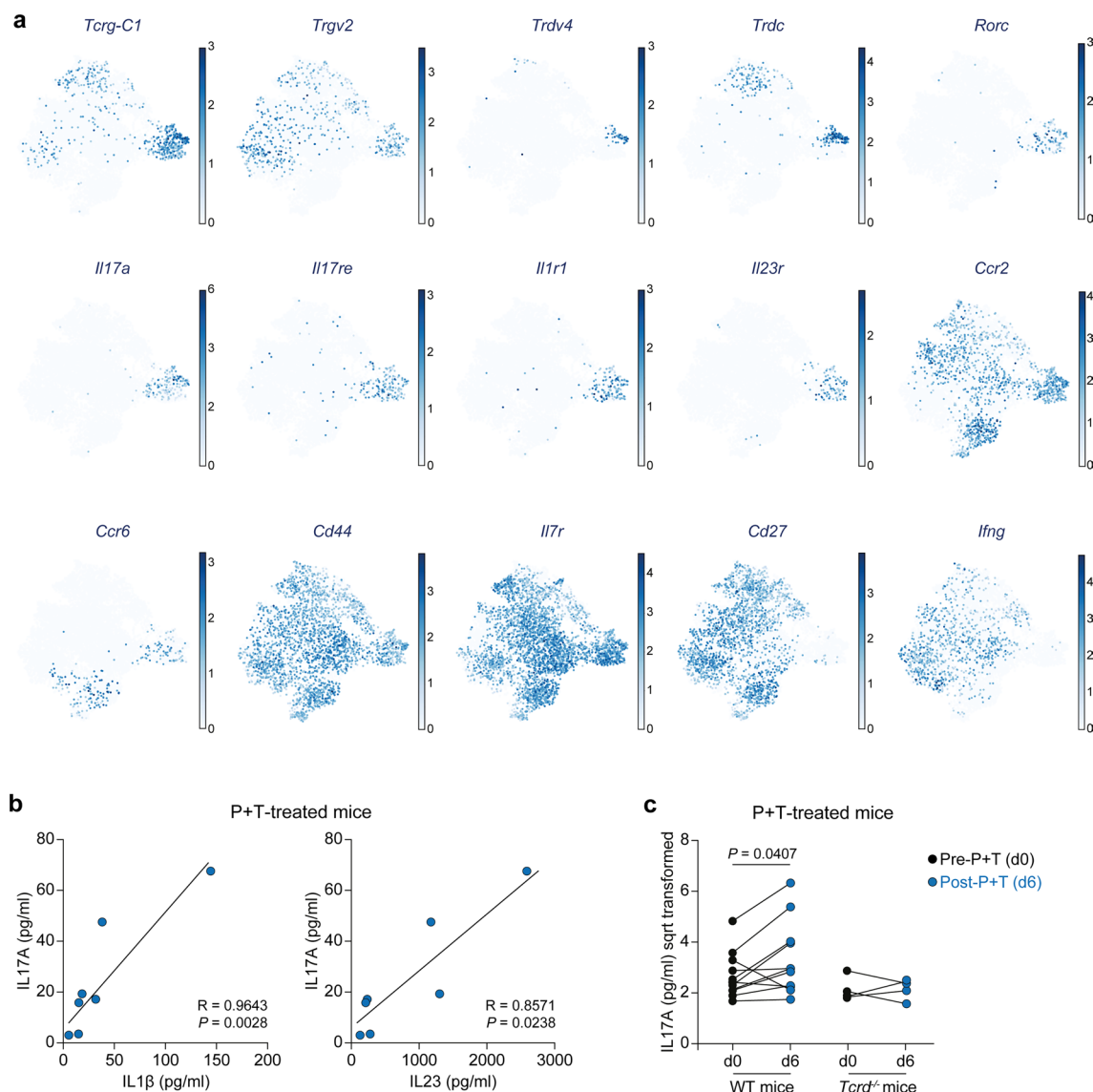


Extended Data Fig. 1 | See next page for caption.

Extended Data Fig. 1 | Palbociclib plus tamoxifen activity against M/D-driven mammary carcinomas is not modulated by CD4⁺ and CD8⁺ T cell depletion.

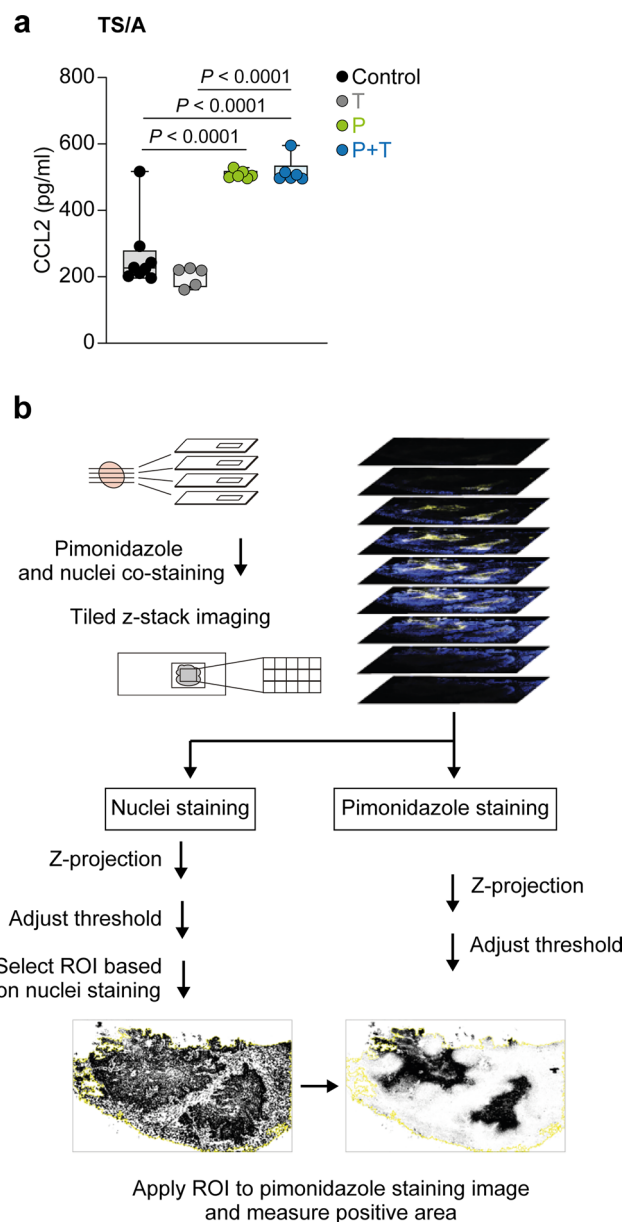
a-f, C57BL/6 J mice bearing M/D-driven tumors were randomly allocated to receive no treatment (control), tamoxifen (T), oral palbociclib (P), focal radiation therapy (RT) followed by (→) T, RT→P (**a-c**), P + T ± α-CD4 and α-CD8 antibodies (**d**), or RT→P + T ± an α-PD-1 antibody (**e, f**). Individual growth curves for primary tumors shown as tumor area relative to the area at the start of treatment (A/A_0) (**a, e**), cumulative relative tumor area (**d**), survival curves (**b, f**), and pie charts with drug responses classified at d20 (**c**), are reported. Pie charts show the percentage of dead mice, objective response (OR; $A/A_0 \leq 0.756$), stable disease (SD; $0.756 < A/A_0 \leq 1.440$), and progressive disease (PD; $A/A_0 > 1.440$), according to RECIST-based criteria. Number of mice, risk ratio (RR) with 95% confidence interval and *P* values are shown. **g**, Bubble heatmap of immune cell

cluster-defining genes by scRNAseq from M/D-driven tumors treated as reported in Fig. 1f. Data show the mean expression of genes in identified cell clusters and the fraction of cells expressing the gene of interest in 40,853 CD45⁺ cells (pooled from 27 tumors, with 3 biological replicates/treatment group obtained from independent mice). **h**, Gini index of TCR γ and δ chain diversity in M/D-driven tumors left untreated ($n = 5$) or treated with P + T ($n = 7$) and harvested at endpoint. Box-plots show means ± interquartile range; whiskers extend to 1.5-fold interquartile ranges. *P* values were determined by linear mixed effects model followed by simultaneous tests of general linear hypotheses (**a, d, e**), two-sided Gehan-Breslow-Wilcoxon test (**b, f**), and two-sided Wilcoxon test (**h**). Part of the results shown in (**b**) and (**f**) are also depicted in Fig. 1b, e (control and RT→P + T), and Figs. 2d, 3f, 4i (control).



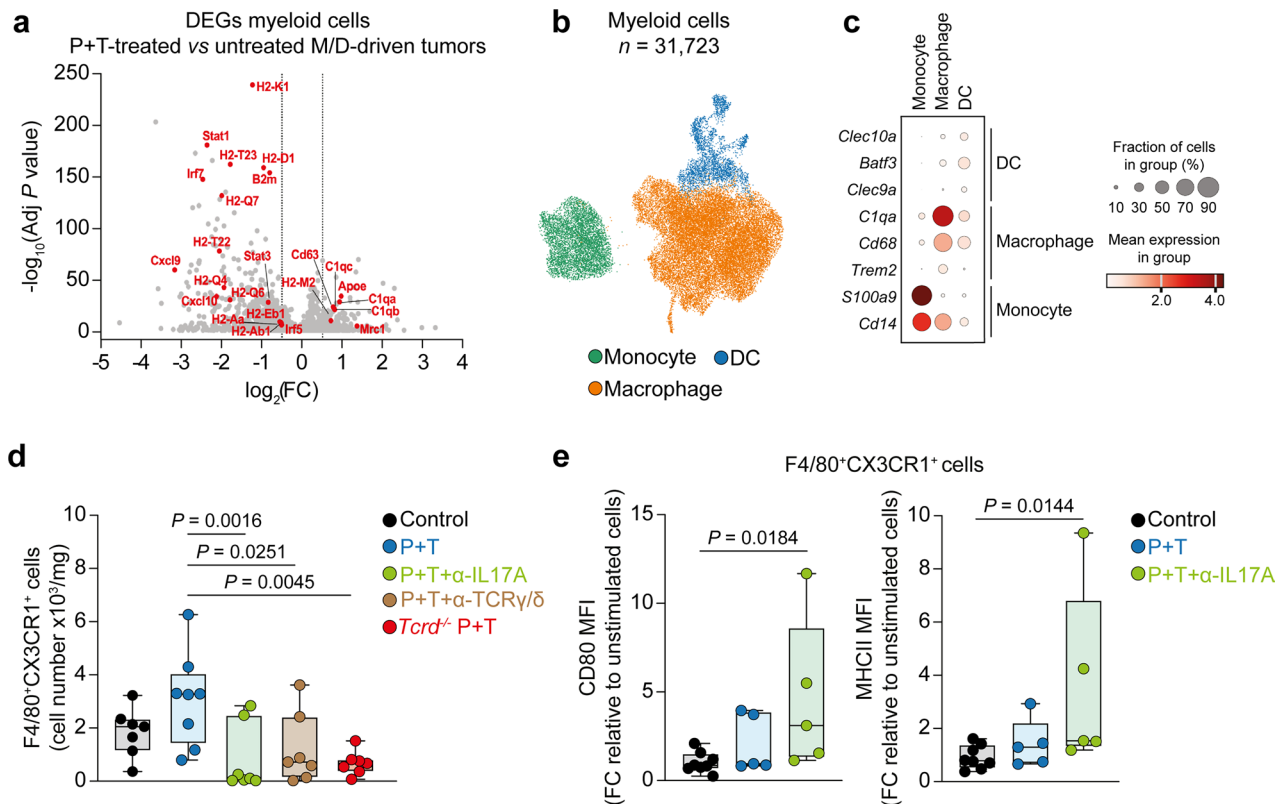
Extended Data Fig. 2 | Transcriptional signature of $\gamma\delta$ T cells recruited to M/D-driven mammary carcinomas upon CDK4/CDK6 inhibition. a, UMAP color-coded for expression of genes associated with the IL17A-producing $\gamma\delta$ T phenotype. Data were obtained from 8,395 T cells (pooled from 27 tumors, with 3 biological replicates/treatment group obtained from independent mice) as reported in Fig. 1f, g. **b**, Correlations between circulating levels of IL17A and IL1 β (left) or IL23 (right) in mice bearing M/D-driven tumors treated with palbociclib

(P) plus tamoxifen (T) on d0-d5 and assessed at d6 ($n = 7$). P (by two-sided Spearman's rank correlation test) and R (Spearman's rank correlation coefficients) values are reported. **c**, Serum levels of IL17A in wild-type (WT; $n = 11$) and *Tcrd*^{-/-} ($n = 4$) mice bearing M/D-driven tumor assessed at d0 (before treatment) and after treatment with oral palbociclib (P) plus tamoxifen (T) at d6. Relevant and significant ($P < 0.05$) statistical differences (by Wilcoxon matched-pairs signed-rank test) are shown.



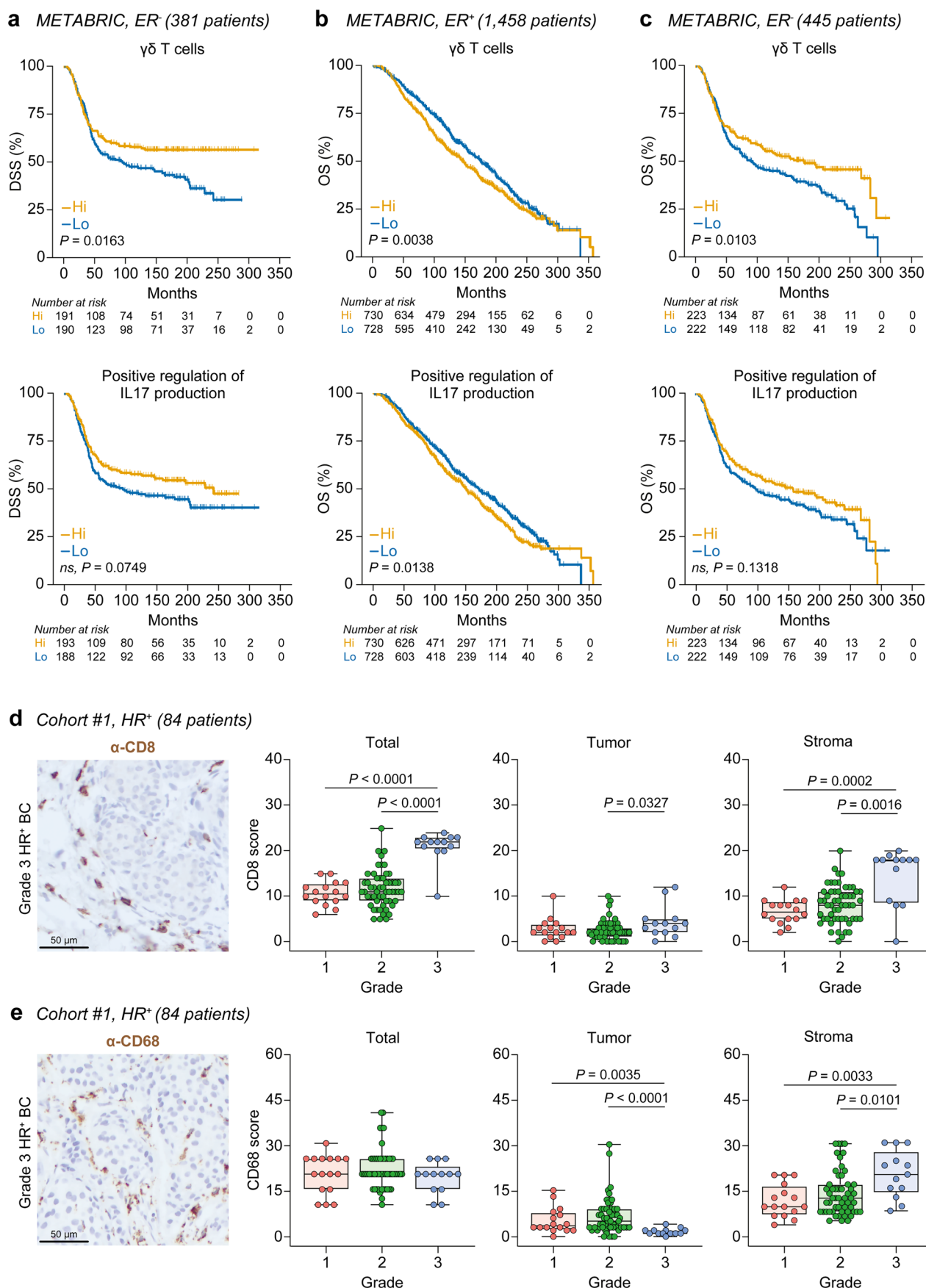
Extended Data Fig. 3 | Impact of CDK4/6 inhibition and hypoxia on CCL2 secretion. **a**, CCL2 secretion from TS/A cells exposed to palbociclib (P), tamoxifen (T) or their combination, for 72 h ($n = 8$ from two independent experiments for control, $n = 5$ from two independent experiments for T, $n = 6$ from two independent experiments for P and P + T). Box-plots overlaid by individual data points show medians $\pm$ interquartile range; whiskers indicate minima and maxima. Relevant and significant ($P < 0.05$) statistical differences (by linear mixed-effects regression followed by simultaneous tests of general linear hypotheses) are shown. **b**, Workflow followed for the quantification of

pimonidazole positive area in M/D-driven tumor tissues. Tiled z-stack images were analyzed with ImageJ. Nuclei and pimonidazole staining images were individually processed with z-projection (sum all slices) and followed by adjusting the threshold in the Otsu method. Regions of interest (ROIs) were selected by auto-detection or manually based on nuclear staining and applied to the pimonidazole staining image to measure pimonidazole-positive area. For representative images and quantification of pimonidazole-positive area, see Fig. 3i.



Extended Data Fig. 4 | CX3CR1⁺ macrophages accumulate in M/D-driven mammary carcinomas exposed to CDK4/6 inhibitors. **a**, Volcano plot of differentially expressed genes (DEGs) in myeloid cells from palbociclib (P) plus tamoxifen (T)-treated ($n = 3$) versus untreated ($n = 3$) M/D-driven tumors. DEGs highlighted in red are discussed in the manuscript and have $[\log_2(\text{FC})] \geq 0.5$ and adjusted (Adj) P value < 0.001 . **b**, UMAP visualization of 31,723 myeloid cells (pooled from 27 tumors, with 3 biological replicates/treatment group obtained from independent mice, as in Fig. 1f) in 3 main phenotypes, as indicated by the color-coded legend (DC, dendritic cell). **c**, Bubble heatmap of myeloid cell cluster-defining genes by scRNAseq from M/D-driven tumors treated as reported in Fig. 1f. Data show the mean expression of genes in identified cell clusters and the fraction of cells expressing the gene of interest in 31,723 myeloid cells (pooled

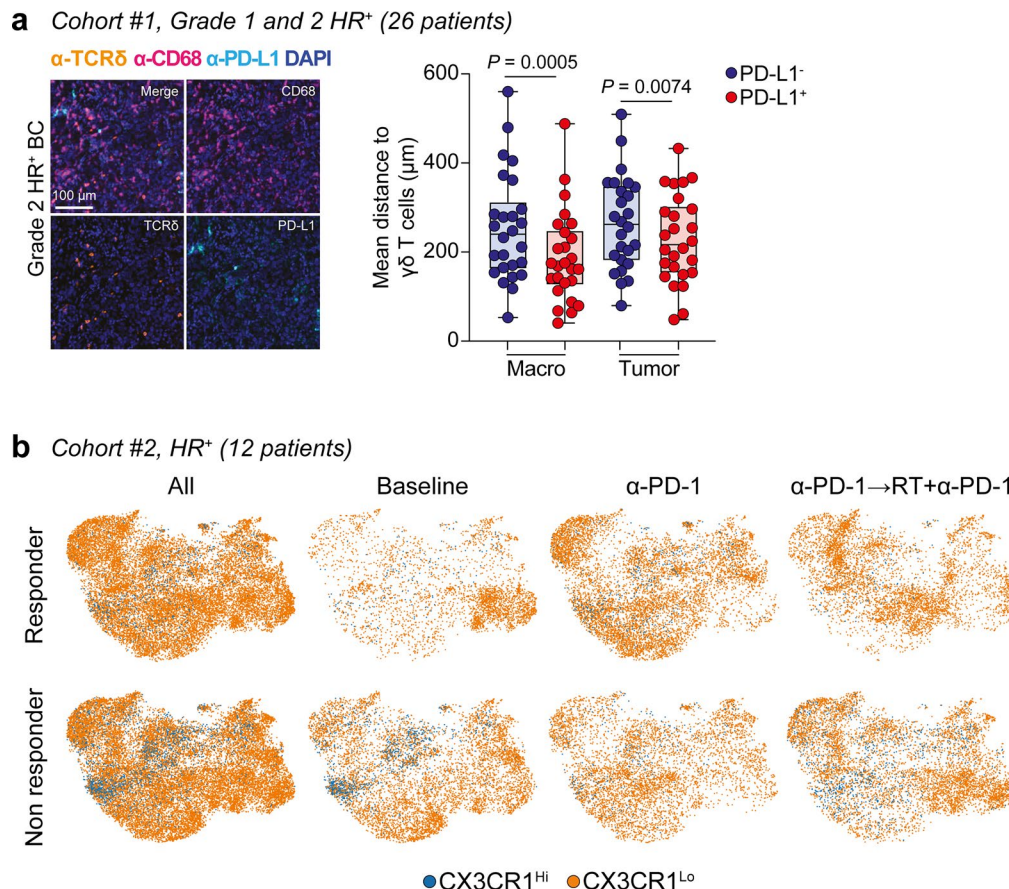
from 27 tumors, with 3 biological replicates/treatment group obtained from independent mice). **d**, Density of F4/80⁺CX3CR1⁺ cells infiltrating M/D-driven tumors established in *Tcrd*^{-/-} mice treated with P+T ($n = 7$) or established in wild-type mice and left untreated ($n = 7$), treated with P+T alone ($n = 8$) or treated with P+T in combination with an α -IL17A ($n = 7$) or an α -TCR γ/δ ($n = 7$) antibody. All tumors were harvested at endpoint. **e**, CD80 and MHCII median fluorescence intensity (MFI) of F4/80⁺CX3CR1⁺ cells infiltrating M/D-driven tumors treated with P+T ($n = 5$), P+T+ α -IL17A ($n = 5$), or left untreated ($n = 8$), harvested at endpoint and subjected to ex-vivo stimulation with LPS/IFN γ . In (d) and (e) box-plots overlaid by individual data points show medians $\pm$ interquartile range; whiskers indicate minima and maxima. Significant ($P < 0.05$) statistical differences (by Kruskal-Wallis with uncorrected Dunn's test) are shown.



Extended Data Fig. 5 | See next page for caption.

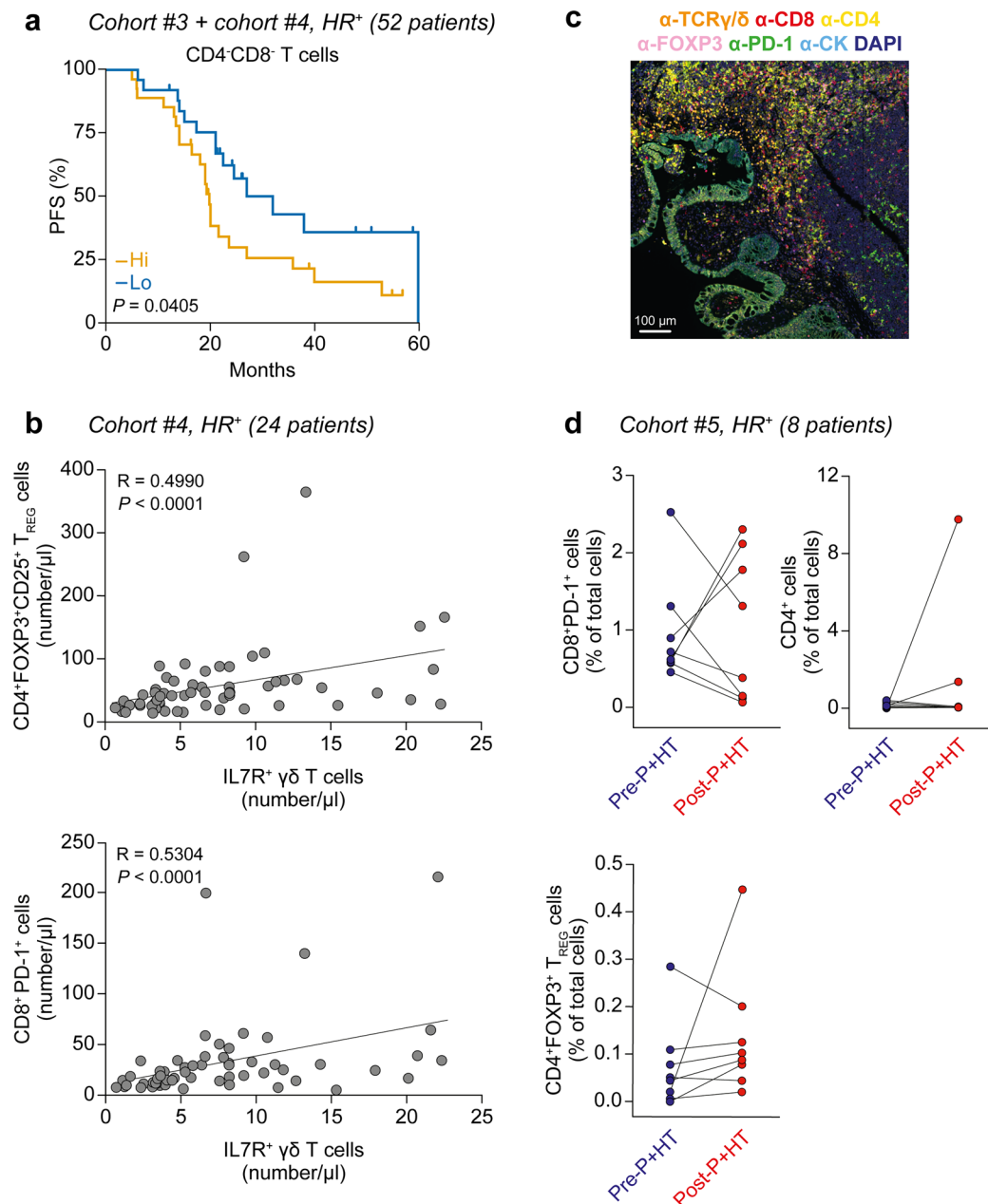
Extended Data Fig. 5 | Impact of $\gamma\delta$ T cells on human ER+ and ER- breast cancer. **a-c**, Disease specific (DSS, **a**) and overall survival (OS, **b, c**) of ER⁺ (**a, c**) and ER⁻ (**b**) METABRIC BC patients stratified by median-single-sample score of $\gamma\delta$ T cell and IL17 signatures in high and low. Number of patients and *P* values (determined by Log-rank test) are reported. **d-e**, Representative images and quantification of CD8 (**d**) and CD68 (**e**) expression by immunohistochemistry in diagnostic

HR⁺HER2⁻ BC biopsies of indicated grade (grade 1: *n* = 16; grade 2: *n* = 55; grade 3: *n* = 13). Box-plots overlaid by individual data points show medians $\pm$ interquartile range; whiskers indicate minima and maxima. Number of patients are indicated. Only significant (*P* < 0.05) statistical differences (by Kruskal-Wallis with uncorrected Dunn's test) are shown. Scale bars, 50 μ m.



Extended Data Fig. 6 | Impact of TAMs on human HR⁺ breast cancer. a, Mean distance of PD-L1⁺/PD-L1⁻ macrophages and tumor cells to the nearest γδ T cell in diagnostic biopsies of HR⁺HER2⁻ BC ($n = 26$) assessed by multispectral imaging. Box-plots overlaid by individual data points show medians $\pm$ interquartile range; whiskers indicate minima and maxima. Relevant and significant ($P < 0.05$) statistical differences (by two-tailed paired t -test) are shown. Scale bar, 100 μ m

b, UMAPs for macrophage clusters by scRNAseq, color-coded for the indicated cell type, from longitudinal HR⁺HER2⁻ BC diagnostic biopsies of responder and non-responder patients receiving preoperative pembrolizumab (α-PD-1) followed by (→) RT plus α-PD-1. Data are pooled from 36 biopsies obtained from 12 patients (3 each: at baseline, after PD-1 and at surgery).



Extended Data Fig. 7 | Levels of circulating and intratumoral CD8⁺PD-1⁺ and T_{REG} cells in HR⁺ BC patients. **a**, Progression-free survival (PFS) curves for patients with HR⁺ BC treated with palbociclib (P) plus hormonotherapy (HT) stratified based on the median abundance of circulating CD4⁺CD8⁺ T cells (at baseline) in high ($\geq$ median; $n = 27$) and low ($<$ median; $n = 25$). Data from patients from cohort #3 ($n = 29$) and cohort #4 ($n = 23$) were pooled. P value determined by Log-rank test is reported. **b**, Correlations between the number of circulating IL7R⁺ $\gamma\delta$ T cells and CD4⁺FOXP3⁺CD25⁺ regulatory T (T_{REG}) cells or CD8⁺PD-1⁺ T cells in longitudinal blood samples ($n = 60$ samples in total) harvested from $n = 24$ HR⁺ BC patients at baseline, after 3 months of treatment and at relapse,

as in Fig. 5h, left panel. P (by two-sided Spearman's rank correlation test) and R (Spearman's rank correlation coefficients) values are reported. **c**, Ulcerative colitis tissue section stained for the simultaneous detection of TCR γ/δ , CD8, CD4, FOXP3, PD-1, pan-CK and nucleus (DAPI) used as positive control for the staining of HR⁺ breast tumors. Scale bar, 100 μ m. For representative images of HR⁺ breast tumors and quantification of $\gamma\delta$ T cells see Fig. 5i. **d**, Quantification of CD8⁺PD-1⁺ T cells, CD4⁺ T cells, and CD4⁺FOXP3⁺ T_{REG} cells in paired HR⁺ tumor biopsies collected at baseline ($n = 8$) and after disease progression on P plus HT ($n = 8$). P values (by two-tailed paired t -test) were all non-significant.

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give <i>P</i> values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	1) Flow cytometry data were collected in (i) MACSQuant Analyzer 10 operated by MACSQuantify v.2.11 (Miltenyi Biotech), or (ii) Attune NxT Flow Cytometer operated by Attune NxT Cytometer Software Version 5.3 (Thermo Fisher Scientific), or (iii) BD FACSCanto II flow cytometry system (Becton Dickinson), (iv) or Fortessa analyzer (BD Biosciences) or (v) CytoFLEX Flow cytometer (BD Biosciences). 2) Pimonidazole-stained images were collected in LSM 880 Confocal Laser Scanning Microscope operated by ZEN Black v. 2.3 SPI (Zeiss). 3) CCL2-protein expression was collected by Immunoblotting on Azure 600® The Ultimate Western Imaging System (Aziure Biosystem). 4) Multiplexed immunofluorescence images were collected in Automated Quantitative Pathology Imaging System (Akoya Biosciences). 5) Immunohistochemistry images were collected in Zeiss Axioscope 5 microscope (Carl Zeiss Microscopy GmbH). 6) Single-cell RNA libraries (mouse and human data) was sequenced on Illumina Novaseq according to the 10X Genomics Single Cell 5' v1 protocol.
Data analysis	1) Flow cytometry data were analyzed with (i) FlowJo v. 9.5.3 (TreeStar, Inc.), or (ii) Cytexpert software (v 2.4), or (iii) LEGENDplex™ Data Analysis Software Suite (https://legendplex.qognit.com; Biolegend). 2) Pimonidazole-stained images were analyzed with ImageJ v1.53t (http://imagej.nih.gov/ij) 3) CCL2-protein expression by Immunoblotting was quantified with Azure software v.1.9.0.0406 (Azure Biosystems). 4) Multiplexed immunofluorescence images were analyzed with (i) inForm software (v2.4.8, Akoya Biosciences), and the phenoptr package [Johnson K (2022). phenoptr: inForm Helper Functions. R package version 0.3.2, https://akoyabio.github.io/phenoptr/], or (ii) QuPath v0.2.3 and Fiji/ImageJ v1.52c. 5) Immunohistochemistry images were analyzed with Histo-scoring system [Detre, 1995, 7490328] 6) Single-cell RNA on human data were analyzed with CellRanger v4.0; CITE-Seq-Count v1.4.3; GMM-Demux; Python 3.8; scvi-tools v0.6.0; cuML v22.02; Scanpy's v1.5.1.

6) Single-cell RNA on mouse data were analyzed with CellRanger v4.0; CITE-Seq-Count v1.4.3; GMM-Demux; Python 3.8; Scanpy's v1.5.1.

Unless otherwise specified in the Methods, data management, analysis and graphing were performed with Prism v. 8.4 (GraphPad) or Excel 365 ProPlus (Microsoft).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

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- Accession codes, unique identifiers, or web links for publicly available datasets
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Data and biological material availability. Mouse scRNAseq and TCRseq data have been deposited in Gene Expression Omnibus (GEO) under accession number #GSE296798 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE296798>). Human scRNAseq data have been deposited as raw count files in GEO under accession number #GSE296805 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE296805>). Human scRNAseq raw fastq files have not been made publicly available due to patient privacy concerns, but can be made available from the corresponding authors upon request. All requests will be fulfilled within two weeks, with data being made available permanently to the requestor. Publicly available transcriptional data from the METABRIC dataset were downloaded from cBioPortal (https://www.cbioportal.org/study/summary?id=brca_metabric) on 11/18/2020. The remaining data supporting the findings of this study are available within the Article, Supplementary Information or Source Data file and/or from the corresponding authors upon request. Source data are provided with this paper.

Research involving human participants, their data, or biological material

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Reporting on sex and gender

Only female patients were involved in these studies. Since breast cancer predominantly affects female, and the biological mechanism we described in this paper is associated with female endocrine physiology, we focus on biological sex in these studies.

Reporting on race, ethnicity, or other socially relevant groupings

N/A

Population characteristics

Cohort #1: 84 women with luminal A (HR+HER2-) breast cancer of different grades [Grade 1 (n = 16), Grade 2 (n = 55) and Grade 3 (n = 13)] range between 32 and 92.

Cohort #2: 12 women with high-risk luminal A HR+HER2- early-stage breast cancer, of which 11 younger than 65 years old, were subjected to preoperative PD-1 inhibition with pembrolizumab, followed by pembrolizumab plus focal RT to the primary breast lesion in 3 fractions of 8 Gy each.

Cohort #3: 29 patients with luminal A and luminal B advanced breast cancer, aged from 30 to 77, received CDK4/6 inhibitors.

Cohort #4: 24 patients with luminal advanced breast cancer aged from 38 to 87 received CDK4/6 inhibitors and endocrine therapy.

Cohort #5: 8 advanced luminal breast cancer patients aged from 42 to 80, received CDK4/6 inhibitors and endocrine therapy.

Recruitment

Cohort #1: patients with luminal A (HR+HER2-) breast cancer were recruited at the Institute of Pathology, University Hospital Halle (Saale), Germany. Biopses were obtained for immunohistochemistry and multispectral immunofluorescence

Cohort #2: Between 2017 and 2021, 12 women with high-risk luminal A HR+HER2- early-stage breast cancer were recruited at the Cedars-Sinai Medical Center (Los Angeles), California on an Institutional Review Board-approved clinical trial of pre-operative pembrolizumab-mediated immune therapy with radiation and serial tissue collection (NCT03366844). Biopses were collected from every patient before treatment initiation (baseline), at 4 weeks, and at 8 months after treatment initiation, for characterization of immune populations by single cell RNAseq analysis during the follow-up of the treatment.

Cohort #3: Between 2017 and 2020, peripheral blood was collected from 29 breast cancer patients that received cyclin-dependent kinase 4 and 6 inhibitors according to local protocols based on both National Comprehensive Cancer Network (NCCN) and ABC International Guidelines, to quantify immune populations by flow cytometry during the follow-up of the treatment.

Cohort #4: Twenty-eight patients with luminal advanced breast cancer were recruited and treated at University Hospital Morales Meseguer (Murcia, Spain). Patients received CDK4/6 inhibitors and endocrine therapy according to local protocols based in NCCN and ABC Guidelines {Cardoso, 2018, 30032243} and were included at the beginning of the treatment in the study HMM-11/18 from June 2018 until May 2021. Serial samples of whole blood in EDTA-K2 10 ml tubes were collected at the beginning of the treatment, at 3 months and at progression of disease, to quantify immune populations by flow cytometry during the follow-up of the treatment.

Cohort #5: Sixty-six patients diagnosed with advanced luminal BC received CDK4/6 inhibitors at the Clínica Universidad de Navarra (Pamplona, Spain). Nineteen patients with matching tumor biopsies obtained before and after CDK4/6 inhibitors were found, but only eight were selected based on tissue quality. Biopsies were collected at baseline and after disease progression on CDK4/6 inhibitors an subjected to multiplex immunofluorescence.

Ethics oversight

All human samples were deidentified, written informed consent was obtained from all participants and the corresponding protocols were approved by the Institutional committees. Specific information is reported below.

Cohort #1: Usage of the anonymized archived biopsy specimen was approved by the local Ethic committee (2017-81) according to the ethical principles included in Declaration of Helsinki 1964 (updated in 2013) (World Medical Association, 2013). Patients' written informed consent was obtained to participate in this study. Luminal B molecular phenotype was determined by immunohistochemistry criteria according to St Gallen International Guidelines{Goldhirsch, 2013, 23917950}

Cohort #2: The samples from this study were used according to the Cedars-Sinai Medical Center. Written informed consent was obtained from all participants and the protocol was approved by the Institutional Review Board (Cedars-Sinai Medical Center; NCT03366844).

Cohort #3: The samples from this study were used according to the Virgen Macarena University Hospital's (Seville, Spain) Institutional Review Board (ref. LCM-INM-2015-01/Law 14/2007, of July 3) according to the ethical principles included in the Declaration of Helsinki 1964 (2013 update). Written informed consent was obtained from all participants.

Cohort #4: The samples from this study were used according to the University Hospital Morales Meseguer (Murcia, Spain). Written informed consent was obtained from all participants and the protocol was approved by the Institutional Review Board (CEIC Hospital Morales Meseguer; EST24/15).

Cohort #5: The samples from this study were used according to the Clínica Universidad de Navarra (Pamplona, Spain). Written informed consent was obtained from patients to participate in this study (ref. 2010.111 mod4) in accordance with the ethical principles contained in the 1964 Declaration of Helsinki (updated 2013) {World Medical Association, 2013, 24141714}.

None of these patients involved in these studies received compensation.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

For in vivo studies, 9-15 mice per group were chosen based on previous experiences, considering the heterogeneity of the model {Buque A et al, 2020, Nat Commun; Petroni G et al, 2021, Clin Cancer Res; Klapp V et al, 2023, J Transl Med}, and following the Care and Use of Laboratory Animals guidelines of Weill Cornell Medical College.

For in vitro and ex vivo experiments, sample sizes were determined based on our previous experience with the specific assay and prior knowledge of expected variation between biological samples in order to achieve meaningful conclusions. Minimal sample sizes were three biological replicates {Klapp V et al, 2023, J Transl Med; Petroni G et al, 2021, Clin Cancer Res; Buque A et al, 2020, Nat Commun; Yamazaki T et al, 2020, Nat Immunol}.

Human sample size was determined by availability of patients and data. No statistical methods were used to predetermine sample sizes, but we included as many patients as possible according to availability for the different analysis.

Sample size is indicated in the figure legends accordingly.

Data exclusions

No exclusion criteria were pre-established in our analysis.

No data were excluded from in vitro experiments. In ex vivo experiments, no data were excluded except for samples with technical problems (i.e., unusual scatter profile, non-specific staining). In in vivo studies, mice who developed extramammary neoplasms (i.e., ovarian carcinoma, thymoma) were excluded.

For single cell RNA-sequencing data, we perform thresholding of cells based on UMI detection (> 30,000), number of unique genes (> 250), total UMIs associated with mitochondrial genes ($\geq 10\%$).

For paired analysis on human samples, patients with missing time points were excluded. For independent analysis on human samples, no data were excluded.

Replication

Unless otherwise indicated, in vitro and ex vivo results were obtained from $n \geq 3$ independent biological samples collected over $n \geq 2$ independent experiments. For in vivo experiments and patient samples, biological replicates are always represented. Single cell RNAseq findings are from a single experiment involving the indicated number of mice. For all the experiments, the exact sample size is reported in the figure legends.

Randomization

Mice were randomly assigned to different groups, except for mice receiving focal radiotherapy, alone or preceding with palbociclib and/or

Randomization	tamoxifen. For those, only animals with a single treatable primary lesion at d0 were selected as the presence of multiple primary non-irradiated tumors at d0 could have complicated the interpretation of the results. For more details on the multifocal nature of the mouse model used in this study see a methodological review published by Buque A and collaborators {Buque A et al, 2021, Methods Cell Biol}.
Blinding	No blinding was performed during in vivo experiments due to the requirement for cage labelling. Blinding was not relevant for in vitro experiments. For single cell and bulk RNA-sequencing studies, investigators who performed data analysis were blinded, as well as for the analysis of human samples and statistical analyses.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

a, Antibodies used for experiments on murine samples
(Antibody Clone Catalogue Number Source Dilution)

Antibodies used for in vivo depletion or neutralization
anti-CCL2 2H5 #BE0185 BioXCell 0.1 mg/mouse
anti-CD4 GK1.5 #BE0003-1 BioXCell 0.4 mg/mouse
anti-CD8 α 2.43 #BE0061 BioXCell 0.2 mg/mouse
anti-CSF1R AFS98 #BE0213 BioXCell 1.2 -> 0.6 mg/mouse
anti-IL17A 17F3 #BE0173 BioXCell 0.1 mg/mouse
anti-PD-1 RMP1-14 #BE0146 BioXCell 0.2 mg/mouse
anti-TCR γ/δ UC7-13D5 #BE0070 BioXCell 0.1 mg/mouse

Antibodies used for flow cytometry
anti-CD45-FITC 30F11 #103122 Biolegend 1:200
anti-CD3 ϵ -PE/Cy7 145-2C11 #100320 Biolegend 1:25
anti-CD8-PerCP/Cy5.5 53-6.7 #100734 Biolegend 1:25
anti-CD4-APC/Cy7 RM4-5 #100526 Biolegend 1:33
anti-TCR γ/δ -APC GL3 #118116 Biolegend 1:77
anti-CD335 (Nkp46)-BV421 29A1.4 #137612 Biolegend 1:50
anti-IL17A-AF488 MF14 #506910 Biolegend 1:200
anti-IFN- γ -BV421 XMG1.2 #505830 Biolegend 1:200
anti-CD80-PE 16-10A1 #104708 Biolegend 1:40
anti-I-A/I-E-PerCP/Cy5.5 M5/114.15.2 #107625 Biolegend 1:625
anti-CX3CR1-PE/Cy7 SA011F11 #149016 Biolegend 1:1250
anti-F4/80-AF647 BM8 #123122 Biolegend 1:200
anti-CD45-APC/Cy7 30-F11 #103116 Biolegend 1:77
anti-CD326-AF647 REA977 #130-117-865 Miltenyi 1:50

Antibodies used for immunofluorescence
RED APC dye-linked IgG1 anti-pimonidazole 4.3.11.3 #HP8 Hypoxyprobe 1:200

Antibodies used for Immunoblotting
anti- β -actin 8H10D10 #3700 Cell Signaling 1:2000
anti-MCP1 (CCL2) 2D8 #MA5-17040 Invitrogen 1:500
HRP-linked anti-mouse IgG N/A #NA931-1ML Cytiva 1:10000

b, Antibodies used for experiments on human samples
(Antibody Clone Catalogue Number Source Dilution)

Antibodies used for immunohistochemistry on samples from cohort #1
anti-CD68 PG-M1 M0876 Agilent - Dako 1:100
anti-CD8 C8/144B M7103 Agilent - Dako 1:50
anti-TCR δ H-41 sc-100289 Santa Cruz Biotechnology 1:100

Antibodies used for multispectral immunofluorescence on samples from cohort #1

anti PD-L1 EIL3N #13684S Cell Signaling 1:200

anti-TCR δ H-41 #sc-100289 Santa Cruz 1:100

anti-CD68 PG-M1 #M087601-2 Agilent 1:150

Antibodies used for multispectral immunofluorescence on samples from cohort #5

anti-TCR $\gamma\delta$ 5A6.E9 #TCR1061 Thermo Fisher 1:100

anti-CD8 C8/144B #GA62361-2 Agilent Ready-to-use

anti-CD4 4B12 #IR649 Agilent Ready-to-use

anti-FOXP3 236A/E7 #ab20034 Abcam 1:300

anti-PD-1 NAT105 #315M Cell Marque Ready-to-use

anti-cytokeratin AE1/AE3 #NCL-L-AE1/AE Leica Biosystems Ready-to-use

Antibodies used for the profiling of circulating T cells by flow cytometry on samples from cohort #3

anti-CD3-APC/Cy7 SK7 #560176 BD Biosciences 1:20

anti-CD4-PE/Cy7 SK3 #557852 BD Biosciences 1:20

anti-CD8-PerCP/Cy5.5 SK1 #565310 BD Biosciences 1:20

Antibodies used for the profiling of circulating T cells by flow cytometry on samples from cohort #4

anti-CD45-BUV395 HI30 #563792 BD Biosciences 1:20

anti-CD3-BUV496 UCHT1 #612940 BD Biosciences 1:20

anti-CD56-BV510 NCAM16.2 #563041 BD Bioscience 1:20

anti-CD4-BV650 OKT4 #317436 Biolegend 1:20

anti-CD8-APC RPA-T8 #301049 Biolegend 1:20

anti-TCR $\gamma\delta$ -FITC B1 #331208 Biolegend 1:20

anti-PD-1-PE/Cy7 EH12.2H7 #329918 Biolegend 1:20

anti-FOXP3-PE 150D #320008 Biolegend 1:20

anti-CD25-BV421 BC96 #302630 Biolegend 1:20

anti-CD11b-PerCP/Cy5.5 ICRF44 #301328 Biolegend 1:20

anti-CD19-PerCP/Cy5.5 HIB19 #982412 Biolegend 1:20

anti-CD127(IL7R α)-AF700 A019D5 #351343 Biolegend 1:20

Validation

All antibodies used are commercially available, validated by the manufacturer and used as listed in Supplementary Table 1.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

Primary mammary carcinoma cell cultures (MGT10 and MGT12) were isolated from Wild-type C57BL/6J female mice of 6-7 weeks of age.
Mouse mammary adenocarcinoma TS/A cells (RRID:CVCL_VQ63) and human mammary adenocarcinoma MCF7 cells (RRID:CVCL_0031) were kindly provided by Dr. Sandra Demaria (Weill Cornell Medicine)

Authentication

Cells were authenticated by STR profiling (a service provided by IDEXX Bioresearch).

Mycoplasma contamination

Cells were routinely checked for Mycoplasma spp. contamination by the PCR-based LookOut[®] Mycoplasma PCR Detection Kit (#MP0035, Millipore Sigma).

Commonly misidentified lines
(See [ICLAC](#) register)

N/A

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

All animal studies were carried out in accordance with the Institutional Animal Care and Use Committee (IACUC) of Weill Cornell Medical College. Wild-type and Tcrd^{-/-} C57BL/6J female mice of 6-7 weeks of age were obtained by Jackson. Mice were housed under standard conditions on a 12 light/12 dark light cycle, 68-77F, and 30% humidity.

Wild animals

No wild animals were used in the study.

Reporting on sex

Only female mice were involved for this study since this study focused on HR+ breast cancer.

Field-collected samples

No field-collected samples were used in the study

Ethics oversight

All in vivo experiments fully complied with protocols approved by the Institutional Animal Care and Use Committee (IACUC) of the Weill Cornell Medical College (n° 2019-2022).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- ☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- ☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- ☒ All plots are contour plots with outliers or pseudocolor plots.
- ☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation

For analysis of tumor-infiltrating immune cells and splenocytes, tumors and spleens were dissociated as described in Methods to obtain single cell suspensions. Cells were labeled with Zombie Aqua™ Fixable Viability kit and blocked with anti-mouse CD16/CD32 (Biolegend). Surface antigen staining was performed in PBS containing 0.5% BSA (FACS buffer). Cells were then fixed with eBioscience™ fixation/permeabilization buffer and further permeabilized with eBioscience™ permeabilization buffer (Thermo Fisher Scientific) according to manufacturer's instructions. For intracellular cytokines analysis, cells were stimulated with phorbol 12-myristate 13-acetate (PMA) plus ionomycin and brefeldin A according to manufacturer's instructions prior to intracellular staining.

For analysis of human circulating immune cells (Cohort #3), whole blood was stained with surface antibodies, and erythrocytes removed as described in Methods. Live cell suspension was resuspended in FACS buffer and analyzed. For analysis of human circulating immune cells (Cohort #4), PBMCs were treated with beriglobine P (CSL Behring) in PBS-based buffer containing 10% fetal calf serum to avoid unspecific staining and labeled with the viability dye PromoFluor 840 (Promocell). Surface antigen staining was performed as described in Methods. Prior to intracellular staining, cells were fixed and permeabilized using True-Nuclear Transcription Factor Buffer Set (Biolegend), following the manufacturer's instructions.

Instrument

For experiments on murine cells, FACSQuant Analyzer 10 operated by MACSQuantify v.2.11 (Miltenyi Biotec) and Attune NxT Flow Cytometer operated by Attune NxT Cytometer Software Version 5.3 (Thermo Fisher Scientific) were used. For experiments on human circulating immune cells (Cohort #3), BD FACSCanto II flow cytometry system (Becton Dickinson) was used. For experiments on human circulating immune cells (Cohort #4), CytoFLEX Flow cytometer (BD Biosciences) was used. For the quantification of serum CCL2 levels, Attune NxT Flow Cytometer operated by Attune NxT Cytometer Software Version 5.3 (Thermo Fisher Scientific) was used.

Software

Data from tumor-infiltrating immune cells and splenocytes, human circulating immune cells (Cohort #3) and human CCL2 serum level were analyzed using FlowJo v. 9.5.3 (TreeStar, Inc.). Data from human circulating immune cells (Cohort #4) were analyzed using Cytexpert software (version 2.4).

Cell population abundance

No sorting was performed for flow cytometry experiments. For scRNAseq, the purity of magnetically sorted CD45+ cells was evaluated by flow cytometry with the appropriate antibodies (CD45 for tumor-infiltrating immune cells). All the isolated cell populations displayed a minimum of 70% purity.

Gating strategy

For flow cytometry experiments on immune cells infiltrating M/D driven tumors, cells were identified using SSC/FSC gate followed by FSC-A/FSC-H to exclude doublets. Single cells were gated on live cells using negative staining for Live/Dead-Aqua. Live cells were further gated on (i) CD45+ leukocytes (see Supplementary Figures 1a and 2) or (ii) on CD3+ T cells (see Supplementary Figure 1b) and then different populations of immune cells based on their expression of specific markers. For flow cytometry experiments on human circulating immune cells (Cohort #3), lymphocytes were identified using SSC/FSC gate followed by FSC-A/FSC-H to exclude doublets. CD3+ cells were identified using positive staining for anti-CD3 antibody and then different populations of immune cells based on their expression of specific markers (see Supplementary Figure 5). For flow cytometry experiments on human circulating immune cells (Cohort #4), lymphocytes were identified using SSC/FSC gate followed by FSC-A/FSC-H to exclude doublets. Single cells were gated on live cells using negative staining for PromoFluor 840. CD45+ cells were identified using positive staining for anti-CD45 antibody and then different populations of immune cells based on their expression of specific markers (see Supplementary Figure 6). Unstained cells, single colors and FMO were used to define positive/negative cell populations. Complete gating strategies for all flow cytometry experiments are shown in the Supplementary information.

- ☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.